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A Survey on the Coding for DNA Storage: Challenges, Algorithms, and Future Directions

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ABSTRACT DNA-based data storage has developed rapidly in recent years and is increasingly viewed as a promising solution for long-term, ultra-dense archival storage. As research moves beyond early demonstrations, encoding has become a central technical challenge that directly affects system reliability, scalability, and practical deployment. This survey presents a structured overview of coding techniques for DNA data storage. We first outline the biochemical and structural constraints of DNA storage, including synthesis and sequencing errors, molecular instability, and context-dependent biases, which strongly influence encoding design. We then review how classical error-correcting codes-such as Reed-Solomon, BCH, convolutional, fountain, and LDPC codes-have been adapted for use in molecular storage systems. Next, we introduce emerging coding approaches designed specifically for DNA storage, including ALD and Damerau distance codes, set-based coding models, and biochemical rule-guided methods such as Yin-Yang coding. Beyond error correction, the survey also examines coding strategies aimed at improving efficiency, with attention to storage rate, biochemical feasibility, and structural robustness. By comparing representative encoding schemes across different channel conditions and application scenarios, this survey summarizes existing theoretical analyses and experimental results and provides practical guidance for selecting suitable codes. Finally, we discuss open challenges and future research directions, highlighting the potential of adaptive, secure, and machine learning-assisted encoding techniques for large-scale DNA data storage systems.

INDEX TERMS DNA data storage, error-correcting codes, conventional coding, advanced coding, bio-constrained coding, efficiency-oriented coding, future coding directions.

I. INTRODUCTION

THE digital revolution has profoundly transformed modern society through data-intensive technologies such as the Internet of Things (IoT), cloud computing, big data analytics, and artificial intelligence (AI). These innovations have led to an exponential increase in global data generation, which is projected to reach hundreds of zettabytes in the coming decades [1]. Traditional storage infrastructures, built on electronic and magnetic media, are increasingly strained by this data deluge due to limitations in scalability, energy efficiency, durability, and cost-effectiveness [2], [3], [4]. These limitations not only present logistical and economic challenges, but also raise concerns about the long-term sustainability of digital preservation.

In response to these challenges, researchers and industry stakeholders have begun exploring next-generation storage paradigms. Among the most promising is Deoxyribonucleic Acid (DNA)-based storage—a biologically inspired medium that offers orders of magnitude improvements in both data density and durability over traditional systems. DNA storage boasts a theoretical capacity of more than 200 petabytes per gram [5], long-term physical stability measurable in centuries or even millennia [6], and exceptional energy efficiency due to its non-volatility and ambient-temperature preservation requirements [7], [8].

Beyond these intrinsic advantages, the application domains of DNA storage can be clearly characterized and modeled, particularly for scenarios where write-once, read-rarely patterns dominate. Typical examples include cold cloud-scale archives, scientific and cultural heritage repositories, medical and genomic databases, and long-term records driven by regulatory or compliance [9], [10]. Consequently, DNA storage is increasingly viewed not only as a conceptual alternative but also as a practical candidate that fits the applications of future archival infrastructures.

Although the immense potential of DNA data storage is well-recognized, the field remains in an early development stage, confronting critical interdisciplinary challenges such as high synthesis costs, limited speeds, error-prone biochemical processes, and—most significantly—a lack of mature encoding algorithms essential for reliable data retrieval. Unlike electronic storage media, DNA exhibits intrinsic biochemical constraints, including alphabet imbalance sensitivity, homopolymer-induced errors, sequence-dependent synthesis and sequencing bias, and stochastic insertion–deletion–substitution noise, all of which necessitate carefully designed encoding techniques to ensure data integrity and scalability. While early pioneering studies, ranging from Joe Davis's artistic Microvenus project [11] to the landmark feasibility demonstration by Church et al. [7], successfully validated basic digital-to-DNA translation, the focus of research has since shifted from simple proof-of-concept to the rigorous optimization of coding strategies tailored to DNA-specific constraints. Addressing the urgent need to mitigate biochemical errors and improve storage density, recent scholarship has expanded the field significantly: Wang et al. [12] provided a comprehensive survey of architectures ranging from direct mappings to fountain codes; Sima et al. [13] targeted the unique landscape of insertion and deletion (indel) errors with custom code constructions and theoretical bounds; and Xiang et al. [14] explored the adaptation of wireless communication codes to molecular channels. Collectively, these efforts highlight that overcoming the physical limitations of the medium relies heavily on the continued interdisciplinary innovation of robust encoding algorithms.

Despite the valuable contributions of existing review works on coding methods, they are often constrained by narrow scopes, with most predominantly emphasizing the direct adaptation of classical digital codes to the DNA domain. They generally overlook the full spectrum of DNA-specific encoding challenges, especially issues such as storage efficiency, biochemical constraints (e.g., GC-content equilibrium, homopolymer suppression), error resilience, and sequence reconstructability. Furthermore, previous surveys rarely integrate core performance metrics and biochemical feasibility criteria into a single unified work, making it difficult to provide comprehensive guidance for the development of practical coding schemes.

To bridge this critical research gap, we adopt a dual-focus strategy: We not only cover established classic coding schemes, as they remain crucial for ensuring the reliability of DNA storage, but also deeply explore emerging, bio-inspired, or constraint-aware coding techniques to broaden research perspectives. This survey allows us to provide a more in-depth and practical analysis than traditional reviews. Our review further investigates the role of key performance determinants—such as coding rate, molecular channel capacity, and biochemical stability—in the field of DNA storage and offers actionable insights for the design of practical, scalable coding frameworks. In this process, we identify several underexplored yet promising research directions, including energy-efficient encoding for low-temperature archival storage, code designs that are random-access-friendly for selective data retrieval, and secure and degradation-resistant encodings to ensure long-term archival longevity.

Accordingly, this paper makes four major contributions.

- It provides a comprehensive introduction to DNA storage that covers the entire pipeline from molecular architecture to system-level operations.
- Details the prevalent error sources in DNA storage and maps them to appropriate mitigation strategies achieved by encoding.
- It surveys both established and emerging coding techniques.
- Provides some possible future research directions.

This work is intended to serve as both an accessible technical reference and a forward-looking roadmap for researchers and practitioners working at the intersection of information theory, molecular biology, and systems engineering. Because one of the primary goals of this survey is to ensure that readers from diverse backgrounds can clearly understand the general principles and design considerations of DNA storage, we intentionally avoid presenting overly detailed or highly specialized code constructions in the main text. Instead, we summarize the essential concepts at an appropriate level of abstraction. By promoting interdisciplinary understanding and innovation in this manner, the authors aimed to help accelerate the transition of DNA storage from laboratory exploration to scalable real-world deployment.

The remainder of this paper is structured as follows. Section II introduces the basic structure of DNA and its relevance to digital information encoding. Section III reviews the core technologies and recent advances that enable DNA data storage, covering synthesis, preservation, and sequencing. Section IV presents key challenges in DNA storage with a focus on error characterization—including substitutions, insertions, and deletions—and mitigation strategies based on coding. Section V discusses fundamental challenges in designing encoding schemes suitable for biochemical, structural, and system-level constraints. Section VI reviews classical error-correcting codes (ECC) adapted for DNA storage, such as Reed–Solomon, Bose–Chaudhuri–Hocquenghem (BCH), convolutional codes, fountain codes and low-density parity-check (LDPC) codes. Section VII examines recently developed codes tailored to DNA-specific error modes, including Damerau metric codes, asymmetric Lee distance (ALD) codes, set-based codes, and Yin–Yang codes. Section VIII highlights advances in efficiency-oriented coding frameworks that account for biochemical properties, molecular constraints, and system throughput requirements. Section IX evaluates the performance of these encoding strategies using theoretical modeling and system-level simulations. Section X outlines future research directions, including adaptive, secure, and machine learning–driven encoding frameworks. Section XI concludes with key insights and emphasizes the central role of encoding design in enabling scalable and practical DNA-based data storage systems.

II. BASIC STRUCTURE OF DNA

A. DNA NUCLEOTIDE STRUCTURE

Nucleotides are the fundamental building blocks of DNA, each consisting of three key components: a deoxyribose sugar, a phosphate group, and a nitrogenous base [15], as illustrated in Fig. 1. The deoxyribose sugar is a five-carbon molecule arranged in a ring structure, with its carbon atoms numbered from 1' to 5′ in a clockwise direction. The phosphate group is attached to the 5′ carbon of the deoxyribose, while the nitrogenous base is attached to the 1' carbon. DNA contains four types of nitrogenous bases: adenine (A) and guanine (G), which are purines characterized by a double-ring structure, and cytosine (C) and thymine (T), which are pyrimidines with a single-ring structure.

B. DNA DOUBLE HELIX STRUCTURE

The iconic double helix structure of DNA and its internal organization are illustrated in Fig. 2. DNA comprises two antiparallel strands twisted into a right-handed helix [16]. Each strand exhibits distinct polarity: the 5′ end terminates with a phosphate group attached to the 5′ carbon of deoxyribose, while the 3′ end features a hydroxyl group (-OH) at the 3′ carbon position. During DNA replication, the enzyme DNA polymerase links new DNA building blocks (nucleotides) to the growing chain. Each new nucleotide connects at two points: its 5′ phosphate group bonds to the 3′ hydroxyl group of the last nucleotide in the chain. This forms the strong backbone of the DNA molecule and allows the strand to grow longer [16] (Fig. 2(a)).

In DNA, the Watson-Crick base pairing rules specify that adenine (A) is paired with thymine (T) via two hydrogen bonds and cytosine (C) is paired with guanine (G) via three hydrogen bonds [16]. This highly specific pairing preserves the double helical structure and ensures the fidelity of DNA replication and transcription. The structural stability of the DNA molecule is further supported by its sugar-phosphate backbone, which forms the outer "sides" of the helix, while the paired bases extend inward to form the "rungs" of the ladder [17], [18] (Fig. 2(b)).

The asymmetric spacing of the backbone creates major and minor grooves: major grooves arise where the backbones are farther apart, and minor grooves occur where they are closer together [19], [20]. These grooves serve as critical binding sites for proteins, which can induce conformational changes in DNA and modulate its function [21], [22]. The antiparallel orientation of the two strands (5′ to 3′ directionality) stabilizes the double helix and facilitates strand separation during replication, allowing each strand to act as a template for complementary synthesis [16] (Fig. 2(c)).

The conformational flexibility of double-helix DNA allows it to adapt structurally to various cellular processes and environmental conditions [23]. Capitalizing on this inherent adaptability, researchers have pioneered DNA-based data storage by encoding binary information into synthetic DNA strands. This approach takes advantage of DNA's natural versatility to achieve ultra-compact, stable, and high-density information storage.

C. DNA ATYPICAL SECONDARY STRUCTURES

Atypical secondary DNA structures refer to non-canonical conformations that diverge from the standard B-form double helix (Fig. 2(c)). These structural variations arise through multiple mechanisms: sequence-specific propensity (e.g. repetitive or GC-rich regions), alterations in physicochemical conditions (including ionic strength, pH, temperature, or mechanical stress), and molecular interactions with proteins or ligands. Such influences can induce structural deviations that include non-Watson-Crick base pairing, alternative backbone conformations, and irregular base stacking arrangements. Representative examples of atypical DNA secondary structures include:

- **Hairpin loops** (also called stem-loop structures) form when a single-stranded DNA region folds back on itself through intramolecular (within the same molecule) base pairing. This structure requires a palindromic sequence (a sequence that reads the same forward and backward) that enables complementary pairing, creating a double-stranded stem region flanking a single-stranded loop. The resulting conformation resembles a hairpin, with the stem maintaining standard Watson-Crick base pairing and the loop containing unpaired nucleotides.
- **Pseudoknots** represent complex RNA/DNA structures where nucleotides in a loop region pair with complementary sequences outside the original stem-loop. The structure contains at least two stem-loop elements with base pairing between them, forming an interlocked three-dimensional configuration. This creates a knotted topology that is more stable than simple stem-loops.
- **Internal loops** (also called bubbles) occur when non-complementary sequences interrupt an otherwise continuous double helix. These features contain unpaired nucleotides on both strands, flanked by normally paired regions, creating a local opening in the DNA duplex. The size can range from 1-2 nucleotides to dozens of bases.
- **Bulge loops** contain extra nucleotides on one strand that have no complementary partners on the opposite strand, causing the extra bases to bulge out of the double helix. These can form due to insertion mutations or during dynamic structural rearrangements, and may range from single nucleotides to longer unpaired segments.
- **Multiloops** (or junction loops) are branching points where three or more double-helical segments meet at a central unpaired region. These complex junctions often

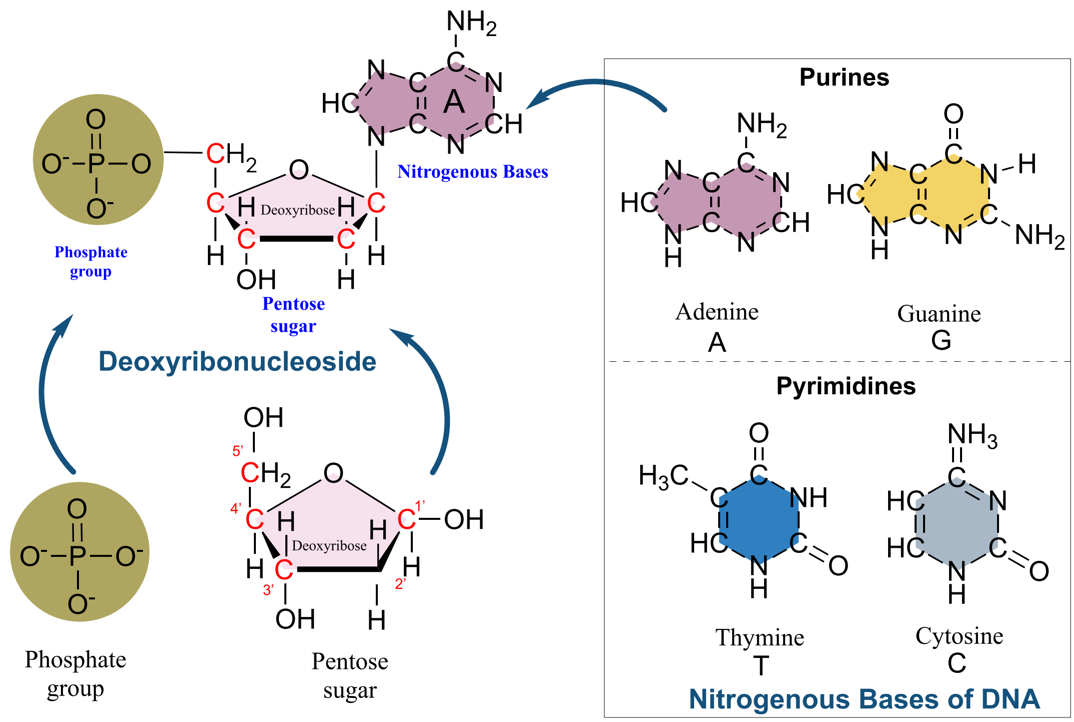

FIGURE 1. Schematic representation of a DNA nucleotide.

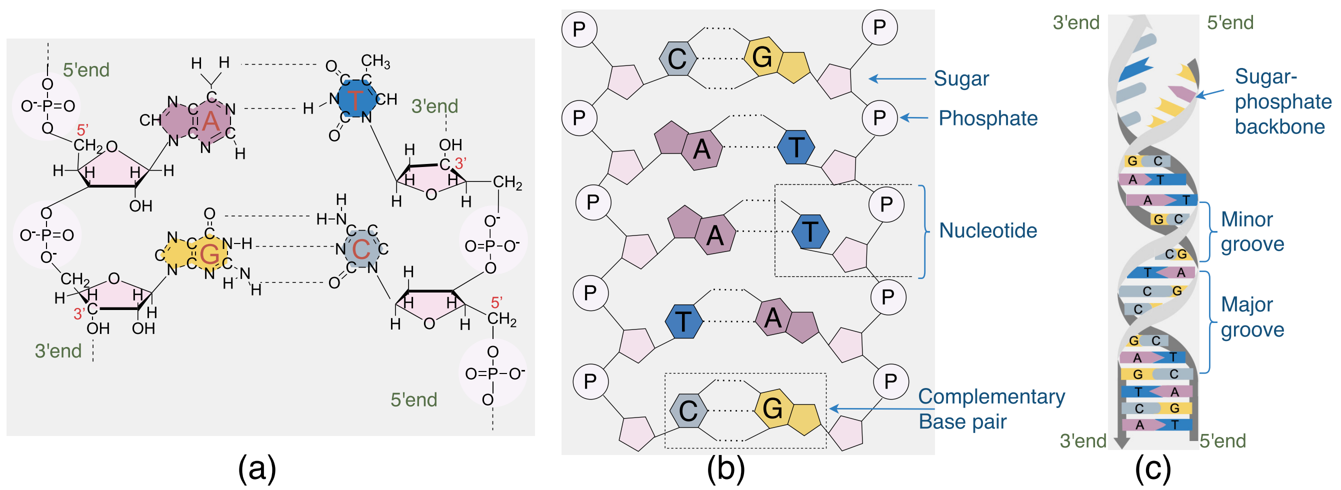

FIGURE 2. DNA double helix structure.

serve as recognition sites for proteins and play important roles in DNA recombination and repair processes [24]. The simplest version is a three-way junction with single-stranded connections between helices.

For structural visualization, Fig. 3(a) schematically represents common DNA secondary structure motifs, while Fig. 3(b) presents a predicted secondary structure for the DNA sequence **S** = GGGAGGACGATGCGGCG-TATTGCGCGAGGATTATCCGCTCATCGTTGTTGTGT-GCAGACGACTCGCCCGA, obtained using the computational prediction tool described in [25].

III. CORE TECHNOLOGIES AND ADVANCES IN DNA DATA STORAGE

The DNA-based data storage process comprises five fundamental steps: encoding, synthesis, storage, retrieval, and decoding. Initially, digital information (including text, images, and audio files) is converted from binary code (0s and 1s) to DNA nucleotide sequences (A, T, C, G) using error-correcting encoding algorithms. The encoded sequences are then synthesized into DNA molecules, with each fragment containing unique addressing information for positional identification. These synthetic DNA molecules are preserved either in cellular systems (in vivo) or in

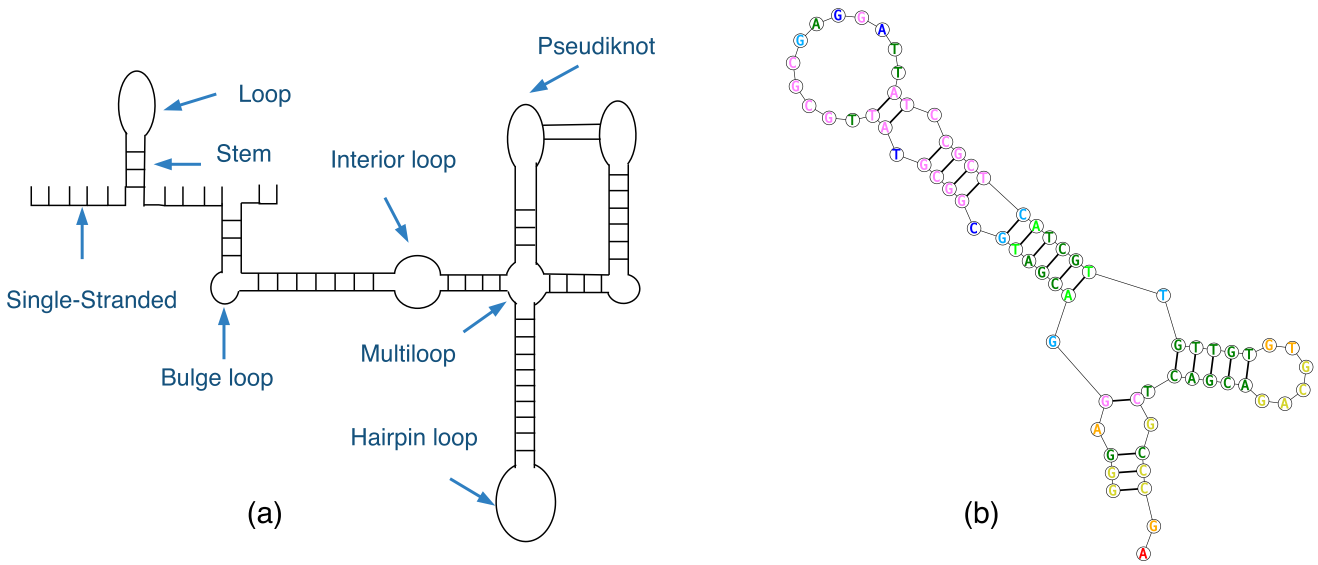

FIGURE 3. DNA secondary structure.

controlled artificial environments (in vitro) [7], with in vitro storage (employing lyophilization or silica-based encapsulation) being the predominant approach for increased stability against environmental degradation [6].

Data retrieval generally utilizes targeted PCR amplification with sequence-specific primers to selectively recover the desired DNA fragments from the storage pool. The amplified products are subjected to high-throughput sequencing (typically using Illumina or Nanopore platforms [26]) to determine the nucleotide sequences. Finally, specialized decoding algorithms reconstruct the original digital information by translating the nucleotide sequences back to binary code.

The subsequent sections detail recent advances in artificial DNA synthesis, preservation techniques, and data retrieval strategies. Fig. 4 provides an overview of the complete DNA data storage workflow.

A. DNA ENCODING TECHNOLOGIES

Traditional DNA data storage relies on encoding digital information into sequences of the four canonical nucleobases (A, T, C, G) [27], [28], offering a theoretical density of 2 bits per nucleotide. Recent studies have demonstrated multiple approaches to expand the DNA alphabet for improved data storage. Tabatabaei et al. [29] achieved an improvement in storage density nearly two times by incorporating seven chemically modified nucleotides alongside the four natural bases. Similarly, Liu et al. [30] utilized 5-methylcytosine (5mC) as a fifth nucleotide while maintaining sequencing compatibility. Zhang et al. [31] developed "epi-bit" encoding that represents binary data through epigenetic modifications of 5mC. In a structural approach, Chen et al. [32] implemented binary encoding using DNA hairpins of different lengths (8bp for "0" and 16bp for "1") attached to a carrier strand.

Despite these advances, key challenges remain: (1) stability issues with modified nucleotides during synthesis and storage, (2) the requirement for advanced sequencing techniques to decode expanded alphabets, and (3) the need for optimized nanopore algorithms to improve decoding accuracy of complex structural encodings.

B. DNA SYNTHESIS TECHNOLOGIES

DNA synthesis technology encompasses both chemical and enzymatic approaches to construct specific DNA sequences, either through de novo synthesis or modification of existing sequences. Chemical synthesis methods have evolved significantly since their inception in the 1950s-1960s [33], progressing from early phosphoramidite chemistry [34] to modern high-throughput solid-phase chip synthesis [35], [36]. While efficient for short oligonucleotides, chemical synthesis faces limitations in producing long sequences because of economic constraints, time-consuming processes, and environmental concerns associated with toxic reagents.

These challenges have driven the development of enzymatic DNA synthesis as a promising alternative, offering advantages in speed, efficiency for longer sequences, and environmental sustainability [37]. Recent breakthroughs in enzymatic synthesis include DNA Script's Syntax platform (99.7% coupling efficiency for 280-nucleotide sequences) [38], [39], the PrimeRoot system for large fragment insertion (up to 11.1 kilobases) [40], and the high-precision Evo model for multi-molecular synthesis [41].

In addition to these synthesis methods, significant progress has been made in DNA assembly technologies. Key techniques include polymerase chain reaction (PCR)-based methods (standard PCR [42], [43], polymerase cycling assembly (PCA) [44], [45], overlap extension PCR (OE-PCR) [46], circular polymerase extension cloning (CPEC) [47]) and advanced assembly approaches (In-Fusion [48],

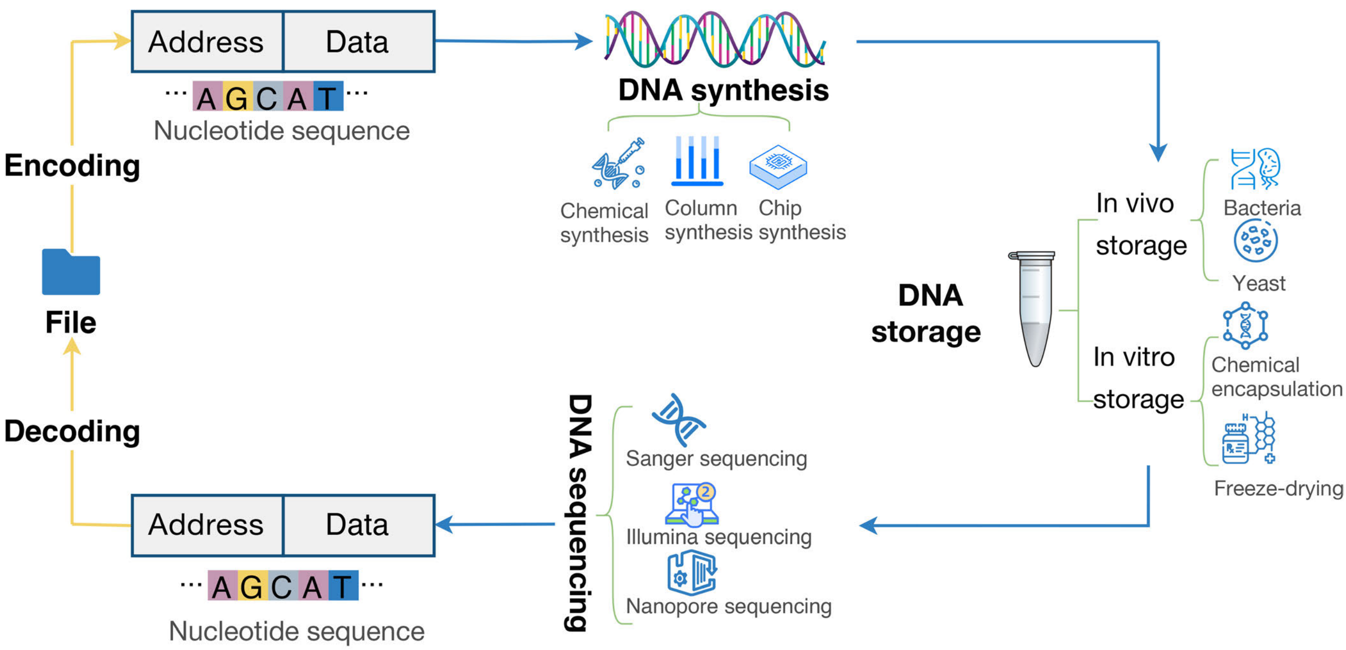

FIGURE 4. DNA data storage workflow.

sequence- and ligation-independent cloning (SLIC) [49], Gibson assembly [50], [51], and Golden Gate assembly [52]).

These technologies collectively enable efficient construction of long DNA sequences while improving ligation and assembly processes, thereby expanding the potential applications of synthetic DNA in various fields.

C. DNA STORAGE TECHNOLOGIES

DNA storage technology is primarily classified into two categories: in vivo and in vitro storage. In vivo DNA storage [53] uses the natural DNA preservation capabilities of living organisms, such as bacteria and yeast. In this approach, data are encoded into DNA sequences and integrated directly into the genomes of organisms'. Their replication mechanisms maintain and can transmit the encoded data over time. The replication and repair processes in living cells provide a more stable and sustainable data storage solution, as they eliminate the need for repeated synthesis or sequencing procedures. For example, CRISPR systems have been used to store digital information by encoding it in genomic DNA [9], [54], [55].

Despite its potential, in vivo DNA storage faces significant challenges that must be addressed. A concern is how exogenous DNA may disrupt the host cell's biological functions or viability. Moreover, extracting and retrieving DNA data requires complex and costly operations, such as cell lysis and sequencing. Additionally, the finite size of cellular genomes constrains storage capacity, and replication errors may lead to gradual loss or corruption of data over time [56].

In contrast, in vitro DNA storage offers alternative preservation methods, including chemical stabilization and silica-based encapsulation [57], [58], while emerging strategies like three-dimensional printed DNA-hydrogel materials offer high-density storage capabilities [59], [60].

A key advantage of in vitro storage is the bypass of intracellular constraints, which enables easier manipulation and protection of DNA. By decoupling storage from living systems, in vitro systems offer tighter control over storage conditions and data longevity.

Consequently, in vitro DNA storage is better suited for large-scale archival long-term preservation applications. However, stability optimization and cost efficiency remain active research challenges.

D. DNA RETRIEVAL TECHNOLOGIES

In DNA-based *in vitro* storage systems, information retrieval is based on two key technologies: Polymerase Chain Reaction (PCR) and DNA sequencing.

1) POLYMERASE CHAIN REACTION (PCR)

Developed by Kary Mullis in 1983, PCR revolutionized molecular biology by enabling the selective amplification of specific DNA sequences *in vitro*. In the context of DNA data storage, PCR plays two critical roles: (1) amplifying encoded DNA fragments during the storage phase and (2) facilitating data retrieval by generating sufficient DNA material for sequencing and decoding. In recognition of its transformative impact, Mullis was awarded the 1993 Nobel Prize in Chemistry [61].

PCR mimics the natural process of DNA replication to exponentially amplify target DNA fragments. Often referred to as a "molecular photocopier," the method can replicate specific DNA segments millions of times in a few hours (Fig. 5). In a typical PCR setup, reaction mixtures are placed in tubes and loaded into a thermal cycler, which orchestrates a series of temperature-controlled steps to drive the amplification process.

Each PCR cycle consists of three main stages:

- **Denaturation (Heating)**: The reaction mixture is heated to 90-95°C (near boiling point), causing double-stranded DNA to separate like an unzipping zipper into two single-stranded templates.
- **Annealing (Cooling)**: When cooled to 50-65°C, artificially designed short DNA fragments (primers) attach precisely to both ends of the target sequence like magnets.
- **Extension (Incubation)**: At 72°C, heat-resistant DNA polymerase (typically the Taq enzyme) synthesizes new complementary strands from the primers, following A-T and C-G pairing rules.

These steps are repeated for 25–35 cycles, with each cycle theoretically doubling the quantity of target DNA. After 30 cycles, this results in an amplification factor of up to 2^{30} (approximately 1 billion copies). This high amplification efficiency enables reliable retrieval of data even from trace amounts of DNA. In practical applications, at least 10 copies per unique DNA sequence are typically required for accurate reconstruction of stored data [62].

To ensure successful amplification, optimal primer design, precise thermal cycling conditions, and appropriate enzyme selection are essential [63], [64].

Modern PCR variants enhance DNA storage applications: Quantitative PCR (qPCR) enables real-time monitoring [65], while Multiplex PCR allows simultaneous amplification of multiple storage fragments [66]. Reverse Transcription PCR (RT-PCR) [67] expands the application to RNA-based storage systems.

2) DNA SEQUENCING

DNA sequencing is a fundamental technique for determining the order of nucleotide bases (A, T, C, and G) on a DNA strand. This process is critical for DNA-based data storage, where digital information is encoded in synthetic DNA molecules. Sequencing technologies have evolved through two generations, each offering improvements in speed, accuracy, and scalability.

a: First-generation sequencing (Sanger method)

Developed by Frederick Sanger in 1977 [68], the Sanger method is based on chain-termination using fluorescently labeled dideoxynucleoside triphosphates (ddNTPs). These ddNTPs lack a $3'$ hydroxyl group, causing DNA polymerase to terminate elongation upon their incorporation. Each ddNTP (ddATP, ddTTP, ddCTP, ddGTP) is tagged with a distinct fluorophore, enabling base identification. After amplification via PCR, the resulting fragments are separated by capillary electrophoresis, and the sequence is determined by their fluorescent signals.

While Sanger sequencing remains the gold standard for accuracy, its low throughput and high cost limit its suitability for large-scale DNA storage applications.

b: Second-generation sequencing (Next-Generation Sequencing, NGS)

NGS has emerged as a key technology for DNA storage retrieval, combining high throughput with cost efficiency [69]. Unlike Sanger sequencing, NGS platforms process millions of DNA fragments in parallel, making them ideal for decoding large datasets encoded in synthetic DNA [70].

Modern NGS platforms fall into two categories:

- **Short-read technologies (e.g., Illumina)**: Use sequencing-by-synthesis to achieve high accuracy ($\sim$99.9%) but are limited to read lengths of $\leq$300 bp [71], [72].
- **Long-read technologies (e.g. Oxford Nanopore, PacBio)**: Employ single-molecule real-time sequencing for reads spanning thousands of bases, facilitating recovery of repetitive or complex sequences [73], [74].

Despite their advantages, NGS platforms face challenges such as homopolymer errors and read-length limitations [75]. To mitigate these, DNA storage systems often integrate error-correcting codes and barcode-based indexing. The following sections explore these coding strategies in detail.

IV. CHALLENGES IN DNA DATA STORAGE: ERROR CHARACTERIZATION AND CODING-BASED MITIGATION

DNA storage systems are inherently vulnerable to multiple types of errors throughout the data lifecycle, including synthesis, storage, and retrieval processes. These errors primarily manifest as substitution, insertion, deletion, and complex compound mutations, each presenting unique challenges for data integrity.

Substitution errors occur when DNA polymerases incorporate incorrect nucleotides (e.g., A→T transversions or C→G transitions). Insertion errors introduce extra bases into the sequence, while deletion errors remove existing bases—both types result in frameshifts that disproportionately affect homopolymer regions. These three fundamental error types are illustrated in Fig. 6(a). In practice, compound mutations—combinations of substitutions and indels (insertions or deletions)—are particularly challenging to detect due to their complex and ambiguous error signatures.

Beyond nucleotide-level errors, PCR amplification introduces significant representation bias in DNA populations [76]. As illustrated in Fig. 6(b), this results in uneven sequence coverage where high-copy fragments dominate while critical low-abundance sequences may be lost entirely. Such biases can compromise the completeness of data recovery.

DNA storage systems face multiple sources of errors including synthesis inaccuracies (both chemical and enzymatic), storage degradation, PCR-induced mutations, and

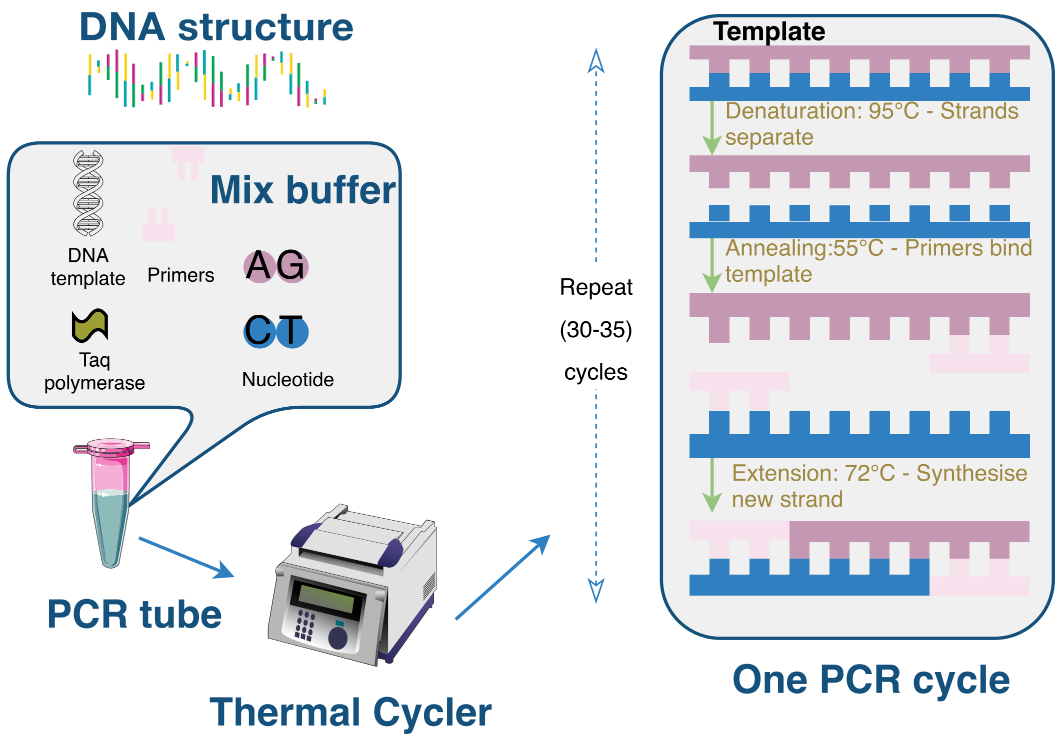

FIGURE 5. Schematic illustration of the Polymerase Chain Reaction (PCR) process.

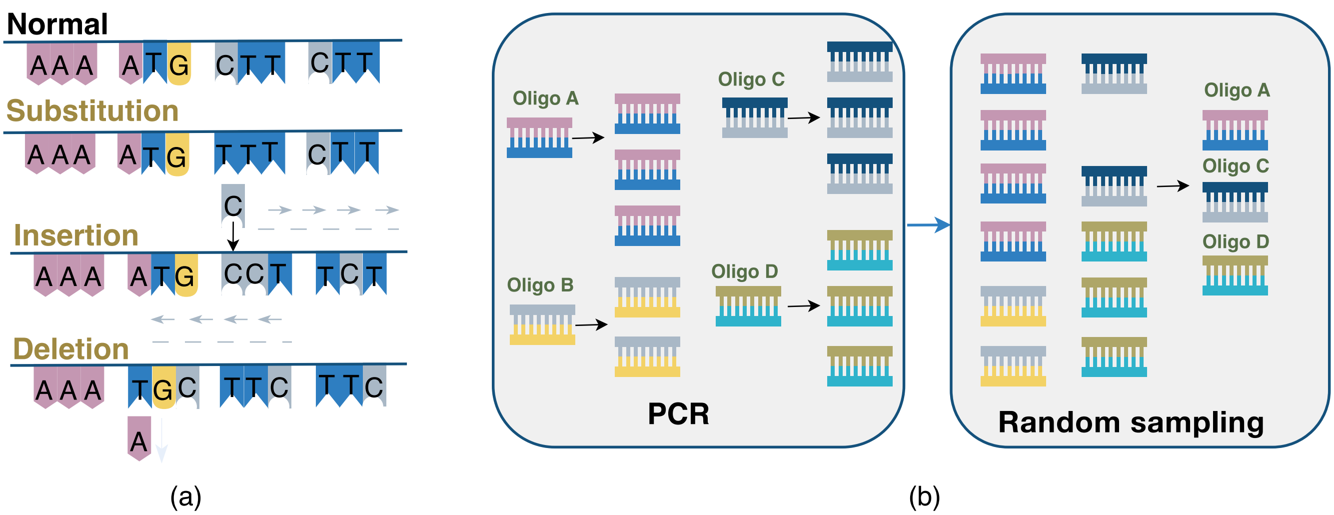

FIGURE 6. Error types in DNA-based storage systems. (a) Base-level errors. (b) Structural-level errors.

sequencing errors, which collectively lead to substitutions, insertions, and deletions. These are compounded by processing challenges such as uneven sequencing coverage and systematic biases [76]. Current mitigation approaches employ high-fidelity synthesis techniques, advanced error-correction codes, optimized storage conditions (temperature/humidity control), and high-throughput sequencing with rigorous quality control. Furthermore, researchers have developed bioinformatics solutions that include bias-correction algorithms [77] and consensus-based statistical methods [78]. While various strategies exist, this work focuses specifically on coding-based approaches for error mitigation in DNA storage systems, which will be detailed in the following sections.

V. CODING CHALLENGES IN DNA STORAGE

The DNA storage coding process (Fig. 7) involves a multi-stage pipeline beginning with source encoding for data compression, followed by channel encoding to introduce error-correction redundancy. The processed binary data are then mapped to DNA nucleotides (A, C, G, and T) using base conversion algorithms, enabling digital-to-biological information translation. For data retrieval, sequencing technologies extract the stored DNA sequences, converting physical redundancy into logical redundancy. The resulting DNA reads then undergo channel decoding for error correction and source decoding for decompression, ultimately reconstructing the original digital information. While this coding framework shows promise, significant challenges remain in algorithm development to optimize storage density, error tolerance, and retrieval accuracy, as highlighted in recent research [12]. The following sections will examine these coding-related challenges and existing solutions in greater detail.

A. BIOCOMPATIBILITY CHALLENGES

Biocompatibility in DNA storage refers to the ability of the designed sequences' to maintain stability while avoiding interference with biological systems. This critical property depends on multiple sequence characteristics including GC content balance, homopolymer avoidance, and reverse complement management.

1) GC BALANCE

GC balance, which refers to the proportion of guanine (G) and cytosine (C) in DNA sequences (typically 40-60% [79]), critically influences DNA stability due to the three hydrogen bonds in the GC pairs versus two in the AT pairs, resulting in higher melting temperatures for the GC-rich regions [80]. However, extreme GC content can promote problematic secondary structures (e.g. hairpins or loops) that compromise data storage reliability. Current solutions employ encoding algorithms to optimize the distribution of GC [81], [82], computational tools for structure prediction, and specialized sequencing techniques for regions rich in GC [83].

2) HOMOPOLYMER

Homopolymers (consecutive identical nucleotides such as "AAAA" or "TTTT") pose significant challenges in DNA storage systems by increasing synthesis errors and causing sequencing inaccuracies due to difficulties in distinguishing identical consecutive bases. These repetitive regions also promote stable secondary structure formation, further complicating data retrieval.

3) REVERSE COMPLEMENT

For any DNA sequence represented as $\mathbf{s} = s_1 s_2 \ldots s_n$ where each s_i denotes one of the four nucleotide bases {A, T, C, G}, three fundamental sequence transformations are particularly relevant for DNA storage systems. The reverse sequence $\mathbf{s}^r$ is created by simply reversing the original base order, while the complement sequence $\mathbf{s}^c$ is formed through substitutions of the base pairing (A$\leftrightarrow$T and C$\leftrightarrow$G). The biologically significant reverse complement $\mathbf{s}^{rc}$ is generated by first complementing each base and then reversing the entire sequence. To illustrate, consider the sequence AGGTC: its reverse version becomes CTGGA, the complement yields TCCAG, and the reverse complement produces GACCT. These reverse complement sequences pose special challenges in DNA storage as they can promote unwanted intramolecular binding through complementary base pairing, leading to the formation of stable secondary structures that interfere with accurate data retrieval [84].

Balancing GC content, avoiding homopolymers, and managing reverse complements are critical for ensuring data stability and reliable retrieval. Proper optimization of these factors helps to maintain the biological inertness of DNA sequences while preserving the integrity of the encoded information.

B. EFFECTIVENESS AND SECURITY CHALLENGES

1) HIGH-DENSITY

High-density encoding represents a fundamental advancement in DNA storage technology, enabling increased data capacity within the same physical volume. By maximizing the information stored per DNA molecule, this approach significantly enhances storage efficiency while simultaneously reducing both spatial requirements and per-bit storage costs. The development of optimized encoding schemes that push the density limits of DNA-based storage has therefore become a primary research focus, as it directly addresses two critical barriers to practical implementation: physical scalability and economic viability. Continued progress in high-density encoding methodologies remains essential for transitioning DNA storage from experimental systems to commercially competitive archival solutions.

2) RANDOM ACCESS

A fundamental challenge in DNA storage systems lies in enabling efficient random access to specific data segments within the massive linear architecture of DNA molecules. Current approaches employ targeted retrieval methods, including magnetic bead separation [85] and PCR-based amplification [86], which provide partial solutions but face limitations in scalability and specificity. The most promising avenue involves developing optimized indexing codes that can precisely identify target sequences without interfering with encoded data payloads. Such indexing systems must balance several competing requirements: they need sufficient uniqueness for accurate targeting, minimal sequence length to preserve storage density, and biochemical compatibility to prevent interference with DNA stability. The development of robust indexing methodologies represents a critical step toward practical DNA storage implementations, as it directly addresses one of the technology's key bottlenecks, the efficient retrieval of specific information from vast molecular databases.

3) SECURITY

Recent studies indicate that DNA storage systems face significant challenges in encryption, integrity protection, and privacy assurance. As a result, security mechanisms related to DNA-based technologies have received growing attention. In the field of multimedia security, DNA-inspired and

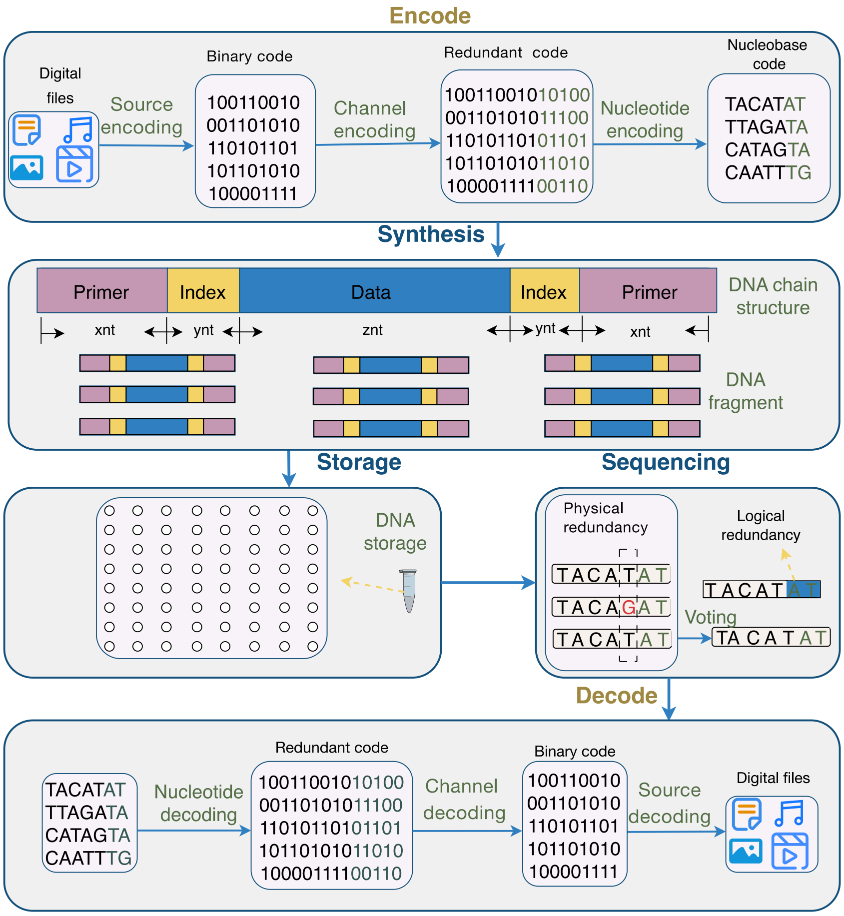

FIGURE 7. Schematic diagram of the DNA storage coding process.

chaos-driven approaches have been explored to enhance confidentiality and robustness. For example, Das et al. proposed a dual-carrier DNA steganography technique [87], while Deb and Bhuyan developed a medical image encryption scheme that integrates chaos with a nonlinear filtering mechanism based on LFSR [88]. More recently, Deb et al. introduced a block-circulant image encryption framework that incorporates DNA–LFSR diffusion [89]. In parallel, regulatory requirements for the secure handling of sensitive molecular information have also emerged [90], [91]. These regulations cover the entire lifecycle of DNA-related data. For example, in frameworks such as GDPR [92] and HIPAA [93], multi-layered protection mechanisms— including cryptography-aware sequence encoding, multi-factor access control, and periodic integrity verification—have been formally established [94].

VI. ADAPTING EXISTING ERROR-CORRECTING CODES FOR DNA STORAGE

In DNA storage systems, conventional channel coding methods play a central role in ensuring that original digital information can be accurately retrieved despite the substantial noise introduced during synthesis, storage, and sequencing. Researchers have adapted classical error-correcting codes—such as repetition codes, Reed–Solomon (RS) codes, Bose–Chaudhuri–Hocquenghem (BCH) codes, convolutional and Turbo codes, low-density parity-check (LDPC) codes, Polar codes, and fountain codes—to counteract typical molecular errors including substitutions, insertions, deletions, and strand dropouts. These codes assist data retrieval by introducing structured redundancy, enabling decoders to detect inconsistencies, correct corrupted bases, reconstruct lost fragments, and statistically infer the most likely original sequences from noisy reads. Together, such coding schemes establish the foundation for reliable end-to-end recovery in DNA-based information storage. In the following sections, we first review these foundational techniques and then discuss their enhanced decoding variants and hybrid architectures, illustrating how these advances further strengthen data integrity throughout the DNA storage lifecycle.

A. EXISTING ENCODING METHODS IN DNA STORAGE

1) REPETITION CODES

Repetition codes serve as a foundational error-correction mechanism in DNA storage by introducing redundancy through bit duplication, enabling the mitigation of mutations,

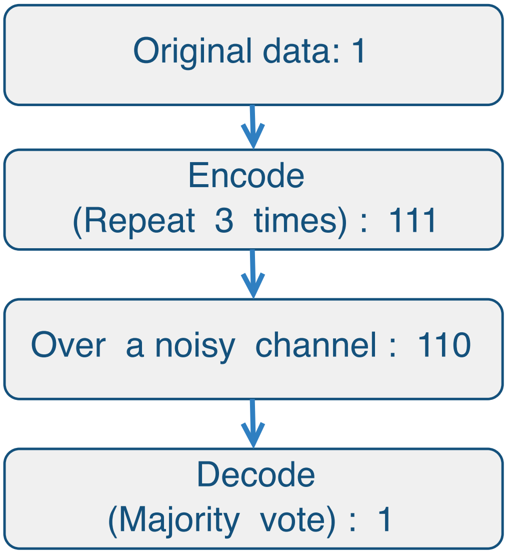

FIGURE 8. Illustration of repetition coding. (1) Encoding: A single data bit (1) is triplicated to create a redundant codeword (111). (2) Transmission: The DNA strand develops a substitution error at the third bit position (111→110). (3) Decoding: Majority voting detects and corrects the error through comparison of all three copies, successfully restoring the original encoded bit value (1).

insertions, and deletions. As illustrated in Fig. 8, this encoding approach facilitates error correction through majority voting schemes. Recent studies [95] have developed two specific algorithms implementing repetition coding to address DNA storage errors, incorporating bioinformatics tools including Basic Local Alignment Search Tool (BLAST) for sequence alignment and Multiple Sequence Comparison by Log-Expectation (MUSCLE) for multiple sequence alignment. These methods quantitatively analyze decoder performance by tracking mutation rates across generations and establishing corresponding bit error rates. The results demonstrate the effectiveness of repetition codes' in enhancing DNA storage reliability against various types of genetic error while maintaining data integrity through systematic redundancy.

2) BCH CODES

BCH codes, a class of cyclic codes renowned for their strong error-correction capabilities, were discovered by Bose, Chaudhuri, and Hocquenghem in 1959 and are named after their inventors [96], [97]. Their encoding and decoding processes, based on finite fields (Galois Fields, GF) [98], involve selecting a generator polynomial, representing information bits as a polynomial, and performing modulo-2 division to generate check bits, which are combined with the original data to form a codeword (see Fig. 9). Recent advances, such as replacing brute-force decoding with syndrome decoding [99], have significantly improved their efficiency, making BCH codes particularly valuable for DNA storage, where minimizing decoding complexity and maximizing reliability are critical for ensuring robust data retrieval and long-term storage stability.

3) REED-SOLOMON (RS) CODES

Reed-Solomon (RS) codes, first introduced by Irving S. Reed and Gideon Solomon in 1960 [100], are a class of powerful error-correcting codes based on polynomial algebra over finite fields. As a special case of BCH codes implemented over Galois Fields $\mathbf{GF}(q)$ where q is typically a power of 2, RS codes maintain the cyclic structure while offering enhanced design flexibility and superior error-correction performance. Recent advances include the VSD (Video Storage in DNA) framework [101], which leverages Reed-Solomon error correction to simultaneously address multiple DNA storage challenges: optimizing information density through video segmentation, ensuring error resilience via robust coding, and maintaining approximately 50% GC content for biochemical stability in synthesized DNA sequences - critical for long-term data preservation.

4) LDPC CODES

Low-Density Parity-Check (LDPC) codes, pioneered by Robert G. Gallager in 1963 [102], represent a class of linear error-correcting codes renowned for their near-capacity performance. These codes are defined by a sparse parity-check matrix $\mathbf{H}$ (a matrix with many zero entries, making the decoding process more efficient), exemplified by this (3,6) regular LDPC code:

$$\mathbf{H} = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix},$$

where the sparse structure enables efficient decoding. As illustrated in Fig. 10, LDPC codes employ iterative belief propagation decoding that effectively handles the error-prone characteristics of DNA storage systems [103], [104]. Table 1 further demonstrates their superiority in DNA storage applications, showing significant improvements in both error correction capability and decoding efficiency.

5) CONVOLUTIONAL CODES

Convolutional codes, first proposed by Peter Elias in 1955 [112], derive their name from the encoding process where each output symbol depends on both current and previous input symbols through a memory mechanism. This temporal dependency makes them particularly effective for communication systems, with detailed encoding/decoding algorithms described by Ivaniš and Drajić [113]. Recent advances have adapted convolutional codes for DNA data storage, including a species-independent (6,3,2) convolutional code design that treats codons as information units while accounting for codon degeneracy [114]. A particularly notable development is the HEDGES (Hash Encoded, Decoded by Greedy Exhaustive Search) code [115], an infinite-constraint-length convolutional code capable of correcting insertions/deletions in single DNA strands without sequence alignment. When paired with outer codes, HEDGES enables petabyte-scale storage with robust error tolerance (up to 10% nucleotide error rates), using a stack algorithm that

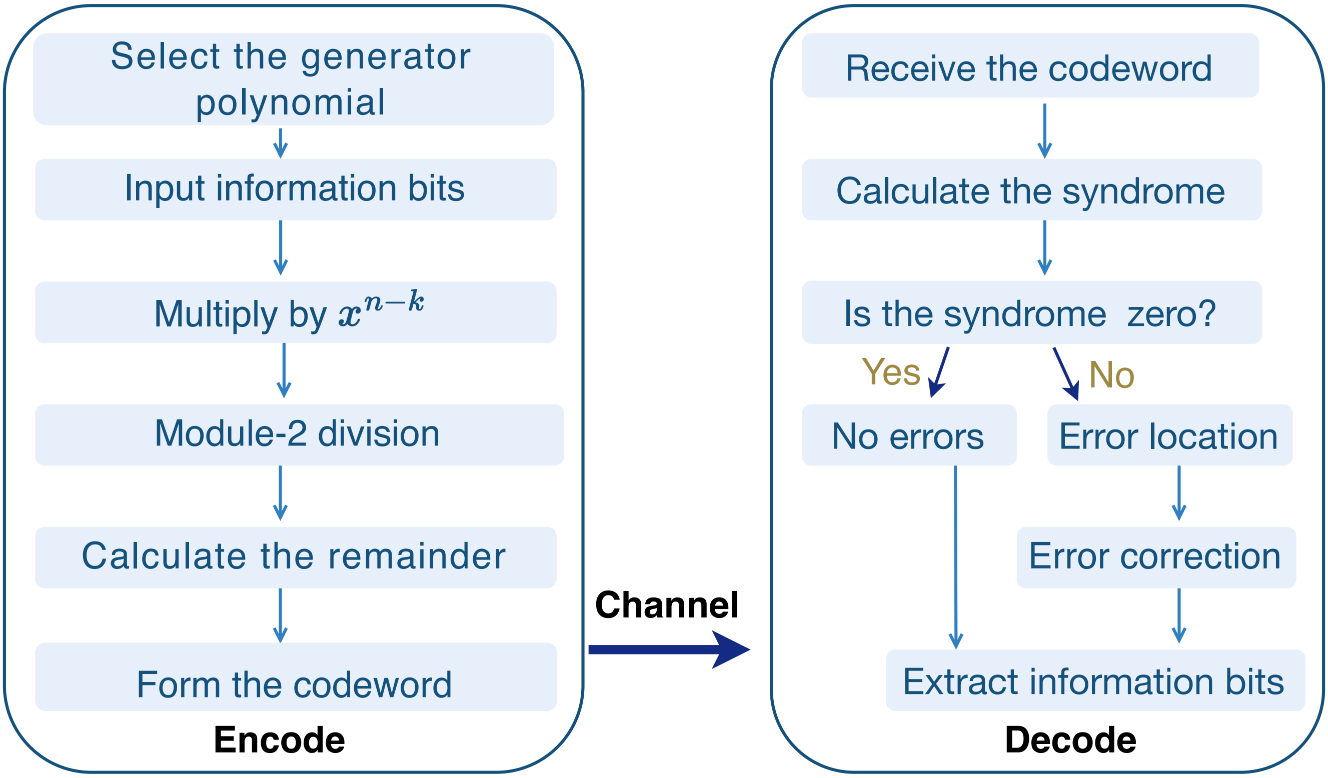

FIGURE 9. BCH code encoding and decoding process. During encoding, a generator polynomial is selected and information bits are represented as a polynomial. This polynomial is multiplied by x^{n-k} (where n is codeword length and k is information length), followed by modulo-2 division to generate check bits. The check bits are appended to the information bits to form the complete codeword. The decoding process involves: (1) syndrome calculation for error detection, (2) error location identification, (3) error correction, and (4) original information bit extraction from the corrected codeword.

either decodes successfully or treats failures as correctable erasures. Further enhancements by Latawa and Aydin [116] have expanded convolutional code applications by proposing several enhancements to existing models in DNA storage, demonstrating their growing importance in addressing DNA sequencing errors and advancing molecular data storage technologies.

6) TURBO CODES

Turbo codes, pioneered by Berrou et al. in 1993 [117], represent a breakthrough in error correction coding by combining multiple convolutional codes with interleavers. This innovative architecture enables near-Shannon-limit performance through iterative decoding, despite increased computational complexity compared to conventional convolutional codes. Their exceptional error-correction capability has established Turbo codes as fundamental components in modern communication systems. The Autoturbo-DNA framework [118] adapts these principles for DNA storage through a PyTorch-based implementation featuring: (1) trainable over-complete autoencoders for error correction, (2) configurable multi-channel noise simulation, and (3) constraint-aware DNA sequence generation. This framework effectively bridges Turbo code theory with the biochemical constraints of DNA data storage while maintaining computational tractability.

7) FOUNTAIN CODES

Fountain codes, pioneered by Luby et al. in 1998 [119], represent a class of rateless erasure codes that excel in unreliable communication channels. Their distinctive feature is the ability to generate an unlimited stream of encoded packets from k original data packets, enabling complete recovery from any subset of $N = k(1 + \varepsilon)(0 < \varepsilon < 1)$ received packets. This elegant property, implemented through LT (Luby Transform) codes [120] and their enhanced variant Raptor codes [121], has found remarkable applications in DNA data storage.

The DNA Fountain approach [122] demonstrates fountain codes' potential by achieving near-theoretical information capacity (215 PB/g) through an efficient encoding scheme that maps binary data to DNA nucleotides (00 $\rightarrow$ A, 01 $\rightarrow$ C, 10 $\rightarrow$ G, 11 $\rightarrow$ T) as illustrated in Fig. 11. This method successfully encoded diverse digital content including a full operating system and video files into 2.14MB of DNA oligonucleotides. Further optimizations emerged through the NOREC4DNA framework [123], which adapts near-optimal rateless erasure codes to DNA's biochemical constraints while minimizing overhead.

Recent advances combine fountain codes with complementary error correction techniques. Jeong et al. [124] integrated fountain codes with Reed-Solomon correction and quality-based sequencing, enabling robust retrieval of 513.6KB datasets from Illumina sequencing. The DR4DNA

TABLE 1. Low-density parity-check (LDPC) code variants for DNA storage systems.

Ref.	Algorithm	Parameters	Performance
[104]	High-rate LDPC with consensus and synchronization mechanisms	Sequence length $N = 256$ bits, code rate $R = 3/4$, variable node degree $d_v = 3$, check node degree $d_c = 12$	Effective error correction under realistic channel impairments
[105]	Interleaved non-binary LDPC for Insertion-Deletion-Substitution (IDS) errors	Block size 4096×100, code rate $R = 3/4$, insertion/deletion probability 2.1×10^{-2}, substitution probability 1.0×10^{-2}	Robust error correction for synchronization issues
[106]	LDPC with heuristic techniques for insertion/deletion errors	Block size 256 kilobits, synthesis/sequencing error rate 4%	Improved trade-off between read-write costs and error correction
[107]	LDPC optimized for erasure and substitution errors	Substitution error rate 4%, variable redundancy and coverage	Reduced reading costs while maintaining reliability
[108]	Asymmetric-Error-Aware Belief Propagation (ABP) decoding	LDPC codes $(500, 1000)$ and $(4096, 4544)$, maximum 100 iterations	Enhanced decoding for nanopore sequencing errors
[109]	Differential Evolution (DE)-optimized LDPC with interleaving	Base size 4×40, code length 17,920 bits, code rate $R = 0.9$	Reduced oligonucleotide reads by 26.25%-38.5%
[110]	Marker-LDPC codes with quadratic encoding	LDPC length 96 bits ($R = 1/2$) + marker 48 bits, overall $R = 1/3$	Effective correction of IDS errors

toolkit [125] represents the latest innovation, employing advanced fountain code decoders for damaged DNA data recovery while supporting integration with content-specific repair methods. These developments collectively advance DNA storage toward practical, high-density archival solutions while maintaining the inherent reliability of fountain code architectures.

8) POLAR CODES

Polar codes, first introduced by Erdal Arıkan in 2009 [126], represent a revolutionary class of capacity-achieving codes for binary-input memoryless symmetric channels. The key innovation of polar codes is *channel polarization*, which is achieved through the recursive application of the *polar transform*. This transform is based on the 2×2 kernel matrix:

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

This kernel operates analogously to the discrete Fourier transform in signal processing [127], and its recursive application leads to the generator matrix $\mathbf{G}_N$, defined as the Kronecker product:

$$\mathbf{G}_N = \mathbf{B}_N \cdot \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{\otimes n},$$

where $N = 2^n$ is the code length, $\otimes$ denotes the Kronecker product, and $\mathbf{B}_N$ is an $N \times N$ bit-reversal permutation matrix [126].

The polarization process transforms N identical channels into a set of *polarized channels*, which are either nearly noise-free or nearly useless. As $N \to \infty$, the fraction of reliable channels approaches the channel capacity. Information bits are transmitted over the reliable channels, while the unreliable ones carry *frozen bits*, typically fixed to zero.

The selection of reliable channels, known as the *information set*, depends on the underlying channel statistics and is central to the construction and performance of polar codes.

For DNA storage applications, *polar codes* can be naturally extended to larger finite fields, such as $\mathbf{GF}(p)$ and $\mathbf{GF}(p^r)$, where p is a prime number and r is a positive integer. These generalized polar codes retain their *capacity-achieving properties* for non-binary input channels, making them well-suited for applications beyond binary systems [128], [129], [130].

In DNA storage systems, where substitution errors are the predominant error type and can be modeled as *memoryless channels*, established polar code construction techniques remain applicable. These include *density evolution* [131], the *Gaussian approximation* method [132], GC-balanced algorithm [133], and *symbol merging algorithms* [134], all of

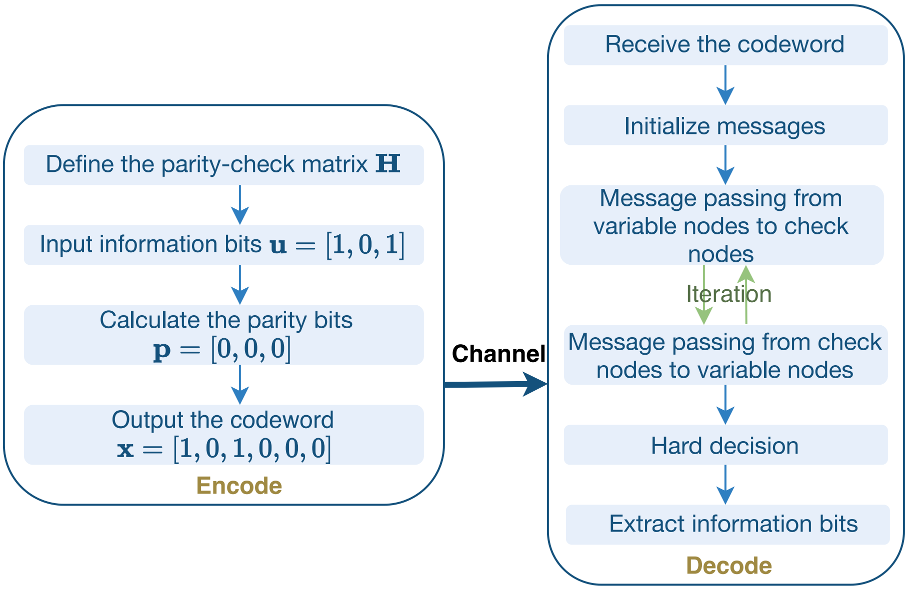

FIGURE 10. LDPC code encoding and decoding process. Encoding transforms information bits u = $[u_1, u_2, u_3]$ into codeword x = $[u_1, u_2, u_3, p_1, p_2, p_3]$ through parity bit generation. Decoding employs belief propagation: (1) Variable nodes initialize using channel outputs, (2) Nodes iteratively exchange probabilistic messages, (3) Decisions are made when either reaching convergence or maximum iterations.

which facilitate efficient and accurate reliability estimation for code construction.

In terms of decoding, *cyclic redundancy check (CRC)-aided successive cancellation list (SCL) decoding* has proven especially effective. This approach significantly improves the error-correcting performance of polar codes, rendering them competitive with more mature schemes such as *LDPC* and *Turbo codes* [135].

More recently, research has extended polar coding techniques to *channels with memory*, which are highly relevant in DNA storage due to the presence of insertions and deletions. Foundational works [136], [137], [138] laid the theoretical groundwork for channel polarization under memory, while follow-up studies [139], [140] investigated *polarization rates* for deletion channels.

On the practical front, several implementations have been developed to support DNA storage in the presence of synchronization errors. Notably, [141], [142], [143] proposed efficient constructions tailored to deletion channels. The recent work by [144] further demonstrated the feasibility of using polar codes to correct *combined insertion, deletion, and substitution errors*—a critical advancement for ensuring robustness and reliability in real-world DNA-based data storage systems.

B. IMPROVING EXISTING DECODING IN DNA STORAGE

In DNA storage research, significant efforts have focused on enhancing decoding algorithms for existing codes like low-density parity-check (LDPC) and fountain codes to enhance performance and address DNA-specific requirements [109], [145], [146], [147], [148]. Notable advancements include Li et al.'s neural network-based decoder for finite-state constrained codes [149], which employs a multilayer perceptron (MLP) to improve error rates by modeling codeword correlations without requiring prior channel knowledge. Further improvements came from a k-order Markov model for DNA channel characterization [150], enabling better error prediction through concatenated codes with convolutional decoders at the cost of increased complexity. Recent work by Ding et al. [151] demonstrated that joint decoding of linear block codes can substantially reduce frame error rates compared to independent decoding. These developments collectively highlight the critical adaptation of traditional coding methods to overcome DNA storage's unique challenges, particularly its distinctive error patterns and biochemical constraints, while maintaining decoding efficiency and reliability.

C. HYBRID EXISTING CODES FOR DNA STORAGE

Recent advances in DNA storage systems have demonstrated the effectiveness of hybrid coding strategies that combine multiple coding techniques to optimize storage density while enhancing error tolerance in high-error environments. A notable approach [152] employs a concatenated scheme integrating Reed-Solomon (RS) codes with Bitwise Majority Alignment (BMA), complemented by data randomization through XOR operations with pseudo-random sequences to prevent repetitive patterns and improve decoding reliability under low-coverage, high-error conditions (including insertions, deletions, and substitutions). Further developments [153] have shown superior performance through innovative concatenations of outer nonbinary LDPC/polar codes with inner convolutional/time-varying block codes, utilizing joint hidden Markov models for multi-sequence symbol

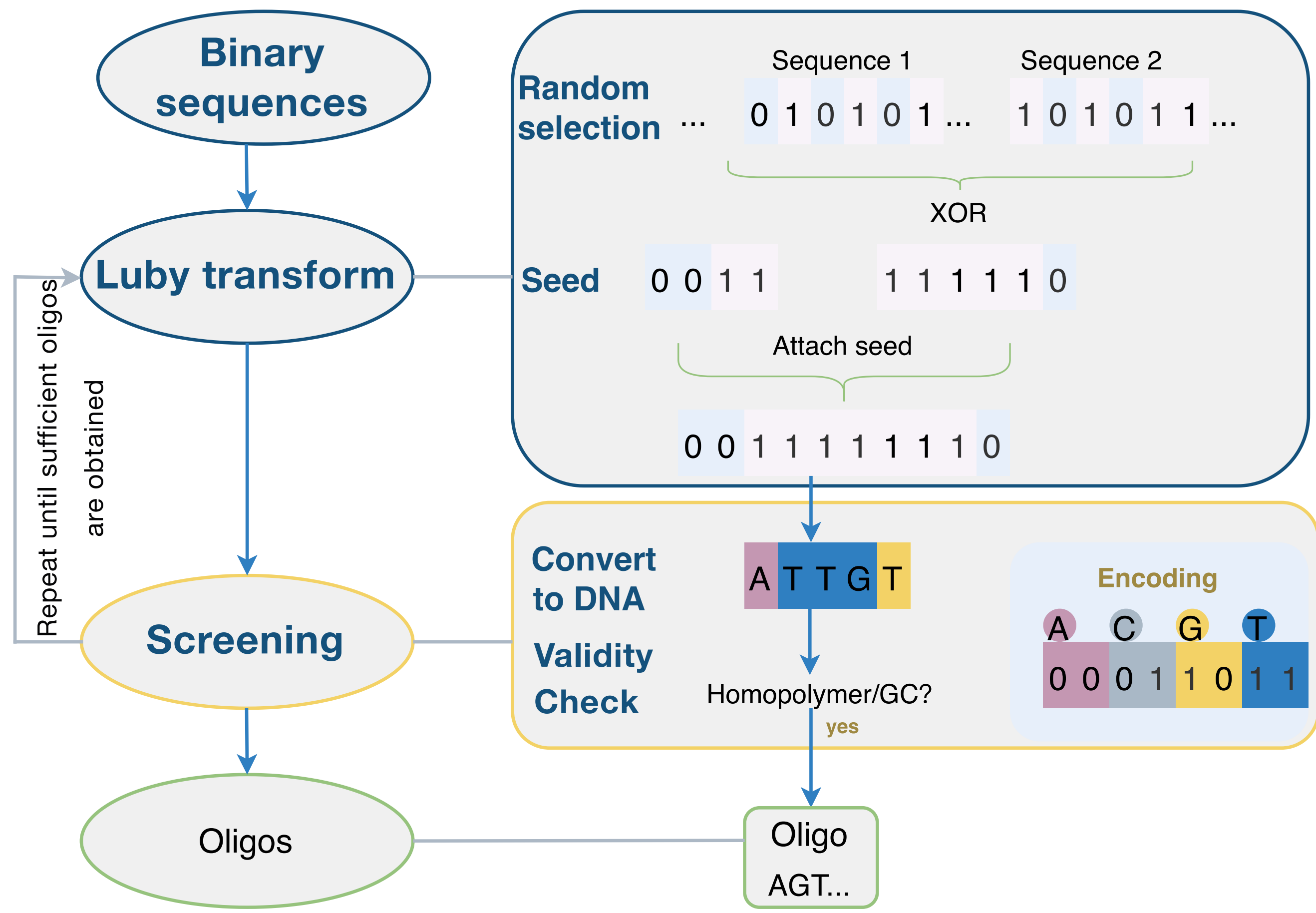

FIGURE 11. DNA Fountain encoding workflow: (1) Binary data segmentation into packets, (2) Seed-based random packet selection for XOR mixing, (3) Seed-payload concatenation, (4) Binary-to-DNA nucleotide mapping (A=00, C=01, G=10, T=11), and (5) Oligo screening.

probability estimation, achieving near-optimal information rates in Monte Carlo simulations. Subsequent optimizations [154] of these concatenated schemes for multi-read scenarios have demonstrated performance closely approaching theoretical achievable information rates (AIR). Most recently, Tong et al. [155] introduced a biologically constrained hybrid system combining marker-assisted LDPC codes with a novel base-symbol synchronization algorithm and improved normalized minimum sum (INMS) decoding, while maintaining critical biochemical parameters ($\leq$250bp strand length, 50% GC content), significantly enhancing both robustness and decoding efficiency for practical DNA storage implementations.

D. INTEGRATING EXISTING CODES WITH BIOLOGICAL CONSTRAINTS

Recent advances in DNA storage systems have demonstrated how integrating constrained coding techniques can effectively eliminate biologically problematic sequence patterns while maintaining compatibility with both current and future technologies. Weindel et al. [156] laid the groundwork for this approach, which was later implemented using bit-insertion-based constrained (BIC) codes combined with rate-compatible LDPC codes [157]. This framework allows for reliable error and erasure correction while enforcing critical biological constraints such as balanced GC content and limited homopolymer lengths, achieving near-channel-capacity information densities. Cao et al. [158] introduced adaptive fountain coding strategies, integrating Reed-Solomon and LT codes with mechanisms to ensure GC content uniformity, nucleotide balance, and sequence continuity, thus improving both storage density and error resilience across diverse environments. More recently, Transfer Coding [159] has emerged as a breakthrough solution for addressing homopolymer constraints, utilizing selective base substitution to achieve near-theoretical performance with minimal sequence modification. Together, these constrained coding approaches address the dual challenge of meeting biochemical requirements and optimizing information density, paving the way for scalable and robust DNA data storage systems.

VII. NOVEL CODES FOR ERROR CORRECTION

Building on the foundations established by classical codes, recent research has increasingly shifted toward coding strategies explicitly designed to satisfy DNA-specific biochemical constraints. This transition reflects a growing recognition

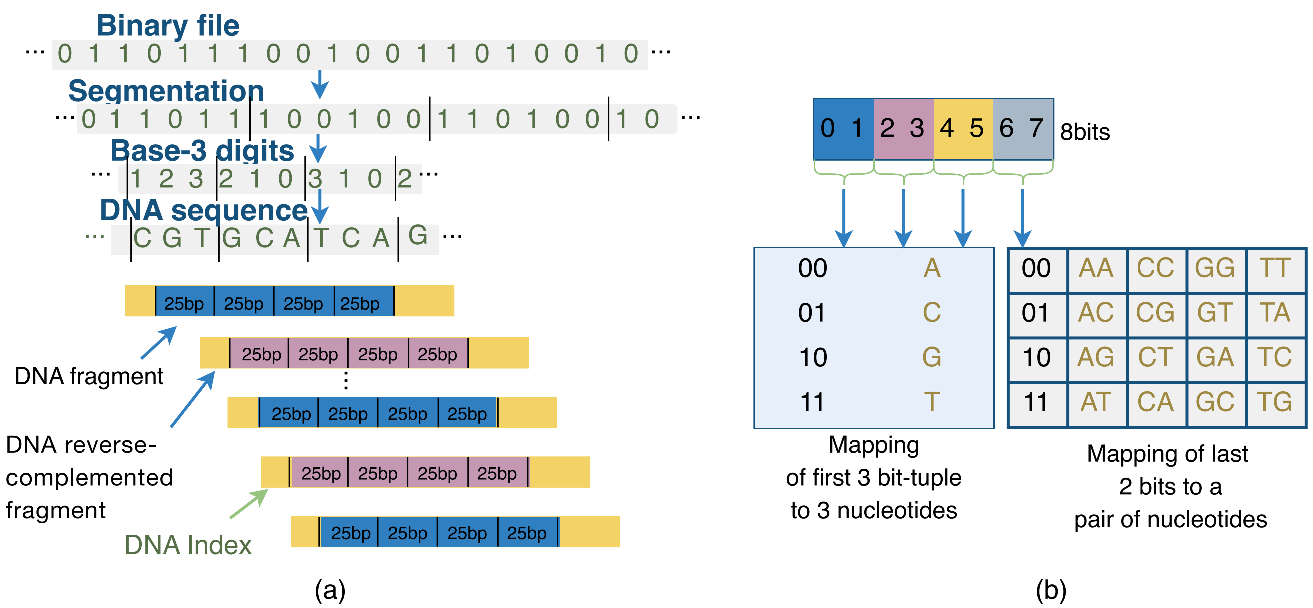

FIGURE 12. Comparative DNA encoding schemes: (1) Goldman's method. (b) Blawat's modulation scheme.

that, while traditional codes provide essential baseline structure, next-generation DNA storage demands coding schemes that intrinsically integrate biochemical feasibility into their design. This section explores several cutting-edge encoding techniques that hold significant promise for advancing DNA storage capabilities.

A. 3-BASE HUFFMAN CODING SYSTEM

Goldman et al. [160] developed an innovative DNA storage architecture employing quaternary Huffman coding for robust binary-to-DNA conversion (Fig. 12(a)). The system implements three key error-control mechanisms: (1) overlapping 100-base fragments with 75-base overlaps creating quadruple sequence redundancy, (2) embedded indexing codes for fragment identification and error detection, and (3) strategic reverse complement operations on alternating fragments. This multi-layered approach provides built-in error correction through redundant sequence coverage while maintaining data integrity during retrieval operations. The design specifically addresses common DNA storage challenges including sequencing errors and strand breaks.

B. TWO-DIMENSIONAL (2D) CUSTOM FORWARD ERROR CORRECTION (FEC) SCHEME

Blawat et al. [161] developed an advanced two-dimensional FEC architecture specifically optimized for DNA storage channels. The system features an efficient 8-bit to 5-nucleotide modulation scheme (Fig. 12(b)) coupled with a comprehensive error protection framework. The protection mechanism integrates BCH codes for address encoding, Reed-Solomon codes for block-level error correction, and CRC codes for error detection, forming a robust defense against various error types. Addressing the characteristic error distribution in DNA strands, the design strategically positions address data at the 5′ end and CRC codes at the 3′ end to mitigate the effects of higher terminal error rates. This spatial organization, combined with interleaved error correction layers, provides systematic protection against both random and systematic errors while maintaining high information density.

C. DAMERAU METRIC CODES

Recent studies have explored DNA sequence disruptions caused by breakages and block deletions, which can lead to various structural alterations [162]. When sequences break at two points, three outcomes are possible: correct reattachment, reverse reattachment, or complete deletion of the broken segment. Less frequently, breaks at three points may cause segment swapping or disintegration. These error patterns have motivated the development of specialized coding schemes that can handle both block transpositions and deletions. Building on the Damerau metric (an extension of Levenshtein distance [163] that considers transpositions), researchers [164] have created DNA codes capable of correcting single deletions/transpositions, later expanding to multiple adjacent transpositions and combined block deletion-transposition errors.

D. ASYMMETRIC LEE DISTANCE (ALD) CODES

The work in [165] introduced Asymmetric Lee Distance (ALD) codes, a novel coding scheme specifically designed for DNA storage systems utilizing nanopore sequencing. These codes address two key challenges: (1) the asymmetric error patterns inherent to nanopore sequencing, and (2) the need for efficient parallel data transmission over DNA's quaternary alphabet. The authors developed a comprehensive framework combining theoretical bounds and practical implementations - using linear programming to derive performance upper bounds while providing explicit code constructions that achieve these bounds. This dual approach resulted in error-correcting codes that are both theoretically sound and practically feasible for DNA storage applications, effectively handling the unique error characteristics of emerging sequencing technologies.

E. SET-BASED CODES

A fundamental challenge in DNA storage systems is the inherent randomness of strand organization - unlike traditional storage media, DNA strands lack fixed physical ordering, making their sequencing order unpredictable during retrieval. To address this disordered nature along with the relatively limited research on coding for insertion/deletion errors, Lenz et al. [166], [167] developed a set-based storage model where data is represented as an unordered collection of M sequences (each length L), accounting for both complete sequence losses and point mutations (insertions, deletions, substitutions). Their work established fundamental limits through Gilbert-Varshamov lower bounds and sphere-packing upper bounds, while developing practical codes approaching these theoretical optima. In parallel, [168] introduced sequence-subset codes with novel distance metrics extending Hamming distance to unordered sets, deriving tight bounds (including Singleton-like and Plotkin-like variants) specifically tailored for DNA storage. Together, these approaches provide a comprehensive framework addressing both the intrinsic disorder of DNA strand organization and complex error patterns, significantly advancing reliable molecular information storage.

F. THE YIN-YANG CODES

The Yin-Yang Codec (YYC) [169] represents an advanced DNA transcoding algorithm that employs complementary encoding strategies: the constrained “Yang” rule ($0 \rightarrow \{A, T\}$, $1 \rightarrow \{G, C\}$) ensures biochemical stability while the flexible “Yin” rule ($0/1 \rightarrow \{A, T, G, C\}$) maximizes information density, together generating 1,536 distinct encoding schemes through combinatorial nucleotide assignments ($C_4^2 = 6$ for “Yang” rules, $4^4 = 256$ for “Yin” rules). As shown in Fig. 13, the process involves binary data segmentation followed by context-aware application of “Yin”/“Yang” rules and majority-voting based nucleotide selection, achieving exceptional recovery rates (99.9% for $> 10,000$ copies, 87.53% for < 100 copies) through its dual-rule architecture that balances sequence constraints with coding efficiency while compensating for sequencing errors via molecular consensus.

G. UNIFIED COMPARATIVE ANALYSIS AND SELECTION OF DNA STORAGE ERROR-CORRECTION CODES

This study conducts a systematic comparative analysis of both classical error-correction codes and recently developed coding schemes. Based on a unified set of evaluation criteria, Table 2 presents a side-by-side comparison of representative coding methods across four key dimensions: error-correction capability, code rate, biochemical compatibility, and decoding complexity. This comprehensive comparison highlights the robustness and low computational burden of traditional block codes (e.g., RS and BCH) in handling base substitution errors, while also revealing the performance potential of modern iterative codes (e.g., LDPC and Polar) in scenarios requiring high capacity and strong error resilience. In addition, DNA-specific coding schemes (e.g., YYC) incorporate biochemical constraints such as GC-balance directly into their structural design, thereby suppressing error generation at the source and compensating for the limited biochemical adaptability of both classical and iterative codes.

Building upon this comparative foundation, Table 3 provides recommendations for selecting error-correction schemes tailored to different channel characteristics and application requirements. These recommendations clarify the correspondence between dominant error modalities and suitable coding solutions. For example, when erasures caused by strand loss or incomplete reading dominate, fountain codes are advantageous; whereas in scenarios where insertions and deletions induce synchronization drift, convolutional codes and polar codes generally offer stronger robustness. Overall, reliable system design typically adopts a cascaded architecture that combines a high-rate, erasure-resistant outer code with an inner coding scheme designed to satisfy biochemical constraints inherent to the nucleotide channel, thereby achieving a balanced trade-off among error-correction capability, computational complexity, and biochemical feasibility.

H. OTHER NOVEL ERROR-CORRECTION SCHEMES

Novel Error-Correction Schemes In recent years, notable advances in coding algorithms for DNA data storage have emerged, including the hybrid HEDGES (Hash Encoded, Decoded by Greedy Exhaustive Search)–Reed-Solomon system for high-rate error correction [115], and optimized DNA sequence design using the CLGBO (Cauchy and Lévy mutation strategy Gradient-Based Optimizer) algorithm [170]. Significant progress has also been made in developing error-resilient encoding schemes, such as linear-time algorithms for edit correction [171], BMVO (Brownian Multi-Verse Optimizer)-optimized thermodynamic coding sets [172], and composite schemes addressing sequencing-depth errors [173]. In addition, innovative approaches like DNA-LC (DNA Levenshtein Code) for multi-strand correction [174] and evolutionary optimization under MFO-RC (Moth Flame Optimizer with Reverse-Complement) constraints [84] further enrich the design landscape.

Building on these foundational efforts, recent breakthroughs also demonstrate substantial improvements in the robustness and efficiency of DNA storage coding. For instance, LDPC codes now support comprehensive error correction—including substitutions, insertions, and

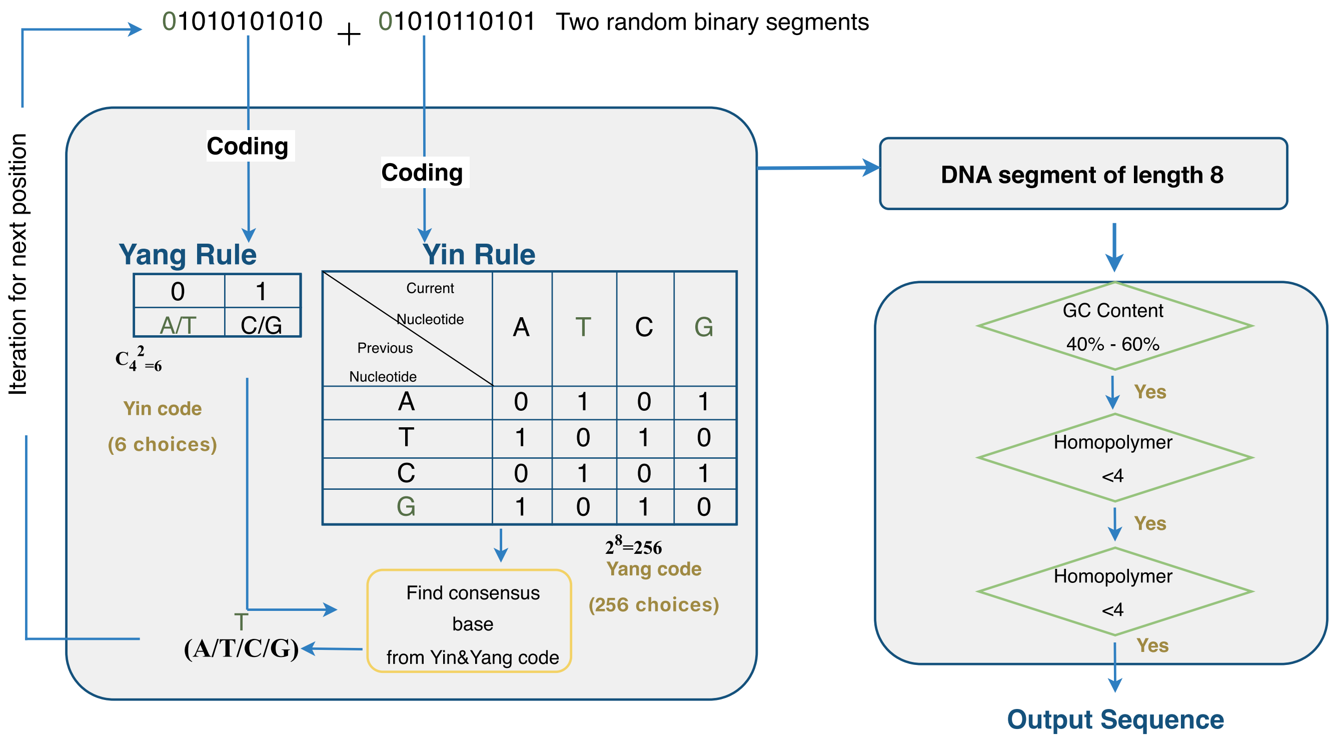

FIGURE 13. Principles of the YYC encoding scheme. Nucleotides shown in green represent symbols used during encoding. The "Yang" rule encodes binary 0 as either A or T, while the "Yin" rule encodes binary 0 as T or G (when preceded by G). The consensus base T is determined through majority voting.

deletions—without requiring additional redundancy or supplementary tools such as erasure codes or synchronization markers [175]. Systematic codes have also shown strong performance in managing random edits while strictly adhering to sequence length constraints [176]. Among the most promising developments is the DNA-QLC scheme, which combines quantized residual variational autoencoder (QRes-VAE) compression with Levenshtein-code-based error correction [177], enabling efficient and accurate DNA-based image storage. Collectively, these innovations offer improved error resilience, enhanced biochemical stability, and greater storage efficiency, significantly advancing the feasibility of practical DNA data storage systems.

VIII. ADVANCES IN EFFICIENCY-ORIENTED CODING UNDER BIOCHEMICAL AND STRUCTURAL CONSTRAINTS

A. UNIFIED OBJECTIVES FOR DNA STORAGE CODING

Coding design in DNA data storage is fundamentally a multi-objective optimization problem driven jointly by biochemical constraints, information-theoretic efficiency, and system-level scalability. To systematically organize these interrelated challenges, this study constructs a unified analytical framework consisting of three core design objectives—efficiency, robustness, and accessibility—which together define the performance limits of practical DNA storage systems.

Efficiency focuses on increasing effective coding density while reducing synthesis and sequencing costs. However, a higher density often raises the risk of forming secondary structures, thereby reducing the synthesis success rates. Robustness requires maintaining the GC balance, avoiding hairpin structures, suppressing insertion/deletion and substitution errors, and ensuring reliable reconstruction of sequence fragments. However, these biochemical and algorithmic constraints typically reduce the usable payload capacity. Accessibility emphasizes random access capability, which depends on scalable and fault-tolerant barcode designs; yet longer or more complex barcodes can reduce synthesis efficiency and introduce additional overhead.

As illustrated in Fig. 14, these three objectives involve both unavoidable trade-offs (e.g., encoding density versus structural stability) and potential positive synergies (e.g., adhering to biochemical constraints can improve synthesis success, while highly fault-tolerant barcodes can reduce indexing errors). This unified perspective indicates that advances in DNA storage coding should not optimize any single metric in isolation but instead seek a balanced solution at the system level, where the objectives are intrinsically coupled. The subsequent sections build upon this unified framework to review recent key developments that address these three objectives while navigating the complex trade-offs inherent in DNA storage coding design.

B. NEW CODING SCHEMES FOR HIGH-EFFICIENCY DNA DATA STORAGE

In DNA storage systems, DNA strand synthesis is a critical step, yet the process is both error-prone and costly. To address

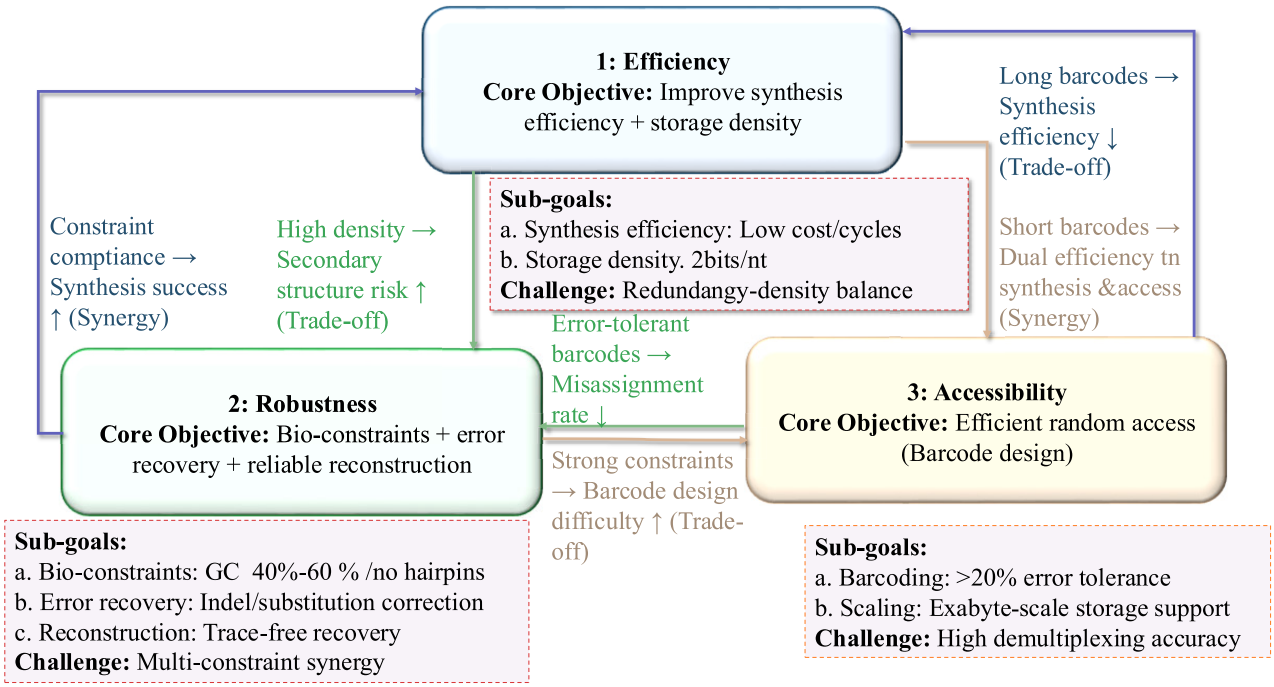

FIGURE 14. A unified conceptual model of the three core objectives in DNA storage coding: efficiency, robustness, and accessibility.

these challenges, researchers have developed a variety of coding algorithms aimed at reducing synthesis errors, shortening synthesis time, improving storage density, and enabling efficient random access to stored data.

1) CODES FOR MINIMIZING SYNTHESIS ERRORS AND TIME

DNA synthesis technologies are mainly based on fixed superscripts to allow parallel construction of multiple strands (Fig. 15) [178]. During this process, the synthesizer iteratively adds nucleotides to selected DNA strands until the superscript sequence is completed. To optimize this distinctive synthesis workflow, several coding strategies have been proposed.

- Lenz et al. [179] introduced a coding method that achieves an information rate of 0.8 bits per cycle by carefully introducing redundancy to balance the synthesis time and the consumption of material. Their approach integrates constrained coding, error-correction mechanisms, and sequence optimization based on finite-state machines, Levenshtein distance, and hierarchical code structures, thus supporting both efficiency and robustness.
- Lu et al. [180] developed two specialized codes targeting synthesis defects—the t-known synthesis defect correction code and the t-synthesis defect correction code. By establishing lower bounds on the redundancy required for correcting a single defect, they demonstrated the near-optimality of their construction, minimizing redundancy overhead to preserve efficiency.
- Immink et al. [181] further explored encoding techniques that transform k source sequences, optimizing synthesis efficiency through precise control of information density, redundancy level, and cycle count.

Together, these works outline a coherent pathway for enhancing synthesis efficiency: improving error control to reduce synthesis failures, reducing cycle counts to boost throughput, and minimizing redundancy to increase effective storage capacity. Together, they provide essential coding foundations for building efficient and reliable DNA storage systems.

2) CODES FOR ENHANCING STORAGE DENSITY

The high cost of DNA synthesis has motivated the development of efficient coding algorithms designed to maximize data storage density—the core of efficiency—while minimizing nucleotide usage. Although DNA offers unparalleled theoretical storage density (up to exabytes per gram), this advantage also introduces challenges in balancing efficiency and robustness (e.g., higher density may increase biochemical or sequencing error risks). Recent innovations have achieved significant progress on both fronts:

- Wang et al. [182] achieved an information density of 1.67 bits per nucleotide—91% of the Shannon limit [122]—using redundancy regulation codes, a 1.05 redundancy factor, 14-bit indexing and 20-bit CRCs. Their design demonstrates an effective balance

between high density and robustness, achieving a low strand-loss rate of only 0.5%.
- Yin et al. [183] enhanced the Harris Hawk Optimization (HHO) algorithm using nonlinear control and adversarial learning to generate DNA sequences that meet run-length, GC-content and Hamming-distance constraints. This approach simultaneously improves storage capacity and robustness through strong constraint compliance.
- Choi et al. [184] introduced degenerate-base encoding, expanding the nucleotide alphabet to 12 characters (A/C/G/T plus 11 degenerate bases). This raises storage efficiency to 3.37 bits per character, representing a major breakthrough in the efficiency dimension. Subsequent work further improved the robustness of degenerate sequences.
- Deng et al. [185] combined variable-length run-length-limited coding with LDPC codes to achieve 1.98 bits per nucleotide, reaching 99% of the theoretical maximum (2 bits/nt) while preserving robustness through effective error correction.

Other important contributions include: Mishra et al. [186], who proposed minimum-variance Huffman coding to enable both lossless and lossy compression for efficiency; Liu et al. [187], who designed mixed-radix coding and achieved near-optimal storage density; Preuss et al. [188], who developed a combinatorial short-sequence–based encoding scheme that achieved a 6.5× increase in density; Wu et al. [189], who developed an end-to-end high-density encoder that enhances full-pipeline performance; and Rasool et al. [190], who integrated opposition-based learning with a moth–flame optimization algorithm to construct stable, constraint-compliant DNA codewords.

In the domain of image storage, Li et al. [191] introduced a hybrid lossy/lossless DNA coding approach that precisely balances storage density and error tolerance; subsequent work [192] incorporated intra-frame prediction from versatile video coding and an enhanced Rubin transform code, achieving a storage efficiency of 1.8 bits per nucleotide; Li et al. [193] further developed digital-pattern-aware DNA coding using rotational codes to improve scenario adaptability; and Sabary et al. [194] employed asymmetric error-correction codes to address the critical issue of short-sequence loss. Collectively, these innovations push storage density ever closer to theoretical limits while maintaining robustness through constraint-aware design.

C. NEW CODING SCHEMES FOR ROBUST DNA DATA STORAGE

1) CODES FOR RECONSTRUCTION

Sequence reconstruction—recovering the original sequence from noisy and fragmented "traces"—is a core challenge to enhance the robustness of DNA coding. Recent studies have made significant progress in reducing the number of traces required and reducing computational complexity while maintaining reliable reconstruction.

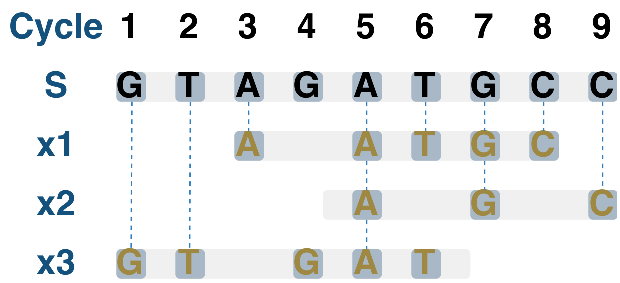

FIGURE 15. Parallel DNA strand synthesis using supersequence S. Three strands are constructed from supersequence S=(GTAGATGCC): (1) x_1 =(AATGC) synthesized at positions {3,5,6,7,8}; (2) x_2 =(AGC) at {5,7,9}; (3) x_3 =(GTGAT) at {1,2,4,5,6}. The complete synthesis requires 9 cycles, corresponding to all distinct positions in S used for these strands.

- Sini and Yaakobi [195] established a theoretical framework for sequence reconstruction, determining the minimum number of traces needed for reliable recovery, thus defining a clear trade-off between robustness (reconstruction accuracy) and efficiency (number of traces).
- Cheraghchi and Gabizon [196] proposed a marker-based coding scheme with redundancy $O(n/\log n)$, which requires an exponential number of traces. This approach illustrates the typical trade-off between efficiency and robustness: reducing redundancy improves efficiency, while using more traces enhances robustness.
- Srinivasavaradhan et al. [197] developed the Trellis BMA algorithm, employing a two-layer structure: the outer code handles sequence loss, while the inner code, based on an improved marker repetition scheme, addresses insertions, deletions, and substitutions. Its linear-time complexity provides scalability advantages for large-scale data applications.

Further innovations have expanded both robustness and efficiency in sequence reconstruction. Pan et al. [198] introduced a two-dimensional molecular system that encodes information in both sequences and backbone structures, simultaneously increasing storage density and reconstruction reliability. Hanna et al. [199] proposed an asymptotically optimal marker-based coding scheme that requires only a constant number of traces for reconstruction, minimizing efficiency costs while ensuring robustness.

2) CODES FOR BIOCHEMICAL CONSTRAINTS

Biochemical constraints—such as reverse complementarity, secondary structures, homopolymer/run-length limits, and GC balance—pose significant challenges to maintaining the robustness of DNA coding. Recent coding schemes have addressed these constraints while maintaining efficiency (by avoiding excessive redundancy) and even supporting accessibility (e.g., generating barcodes that satisfy biochemical constraints).

- **GC balance**: Lu and Kim [200] developed GC-balanced codes using multiple parity formats and bit-flipping

TABLE 2. Comprehensive comparison of classical and advanced encoding methods for DNA storage.

Ref.	Method	Error Correction Capability	Code Rate	Biochemical Compatibility	Decoding Complexity	Representative Performance
I. Classical Encoding Methods						
[94]	Repetition Codes	Corrects substitutions, insertions, deletions (via majority voting)	Low (redundancy)	General	Low	Reduced mutation rates; improved BER across generations
[95]-[97]	BCH Codes	Corrects multiple random errors (Galois Fields)	Medium (configurable)	General	Medium (Syndrome decoding)	Enhanced robust data retrieval; stable long-term storage
[99], [100]	Reed-Solomon (RS)	Corrects burst errors; handles substitutions/erasures	High	Good (≈50% GC in VSD)	Medium	Optimized info density; stable preservation in video storage
[101]-[103], [109]	LDPC Codes	Corrects substitutions, insertions, deletions (iterative BP)	High ($R = 0.9$)	General	Med-High (Iterative)	26.25%–38.5% reduction in oligo reads; robust to IDS
[111]-[114]	Convolutional Codes	Corrects insertions/deletions (temporal memory)	Medium	General	Medium (Stack algo)	10% nucleotide error tolerance; petabyte-scale support
[116], [117]	Turbo Codes	Near-Shannon-limit error correction	High	Good (Constraint-aware)	High	Bridges communication theory with DNA constraints
[118], [121], [123]	Fountain Codes	Corrects erasures/substitutions (from $k(1+\varepsilon)$ packets)	Very High (215 PB/g)	Good (GC-balanced)	Medium	Encoded OS/video; 513.6KB dataset robust retrieval
[125], [143]	Polar Codes	Corrects substitutions, indels (channel polarization)	High	General	Medium (SCL)	Robust to combined indel/sub errors; handles memory channels
II. Advanced Encoding Methods						
[159]	3-Base Huffman	Enhances error correction (overlapping fragments)	High	General	Low	Pioneered DNA storage; improved density
[160]	2D FEC Codes	Corrects synthesis/sequencing errors (8-bit → 5-nt)	Medium	General	Medium	Targeted error mitigation for DNA synthesis
[161]-[163]	Damerau Metric	Corrects adjacent symbol exchanges + block deletions	Medium	General	Medium	First to handle adjacent exchanges in DNA systems
[164]	ALD Codes	Corrects asymmetric errors (nanopore-specific)	Medium	Good (Nanopore tailored)	Medium	Optimized for nanopore error patterns
[165]-[167]	Set-based Codes	Improves error resilience (theoretical bounds)	Medium	General	Medium	Established code performance bounds
[168]	Yin-Yang (YYC)	High recovery rates (biochemical-compatible)	Medium	Excellent	Medium	High data recovery rates via transcoding

TABLE 3. Guideline for selecting error-correction schemes based on channel characteristics and application requirements.

Application Scenario	Dominant Error / Constraint	Recommended Scheme	Rationale & Implementation Strategy
1. Low-Latency / Real-Time Retrieval	Random Substitutions (Bit flips); Low Indel rates	**BCH, RS**	Best for scenarios requiring fast decoding with minimal computational overhead. RS is particularly effective when errors occur in small, predictable bursts.
2. High-Density Archival DNA Storage	Complex noise mixtures; Approaching Shannon Capacity	**LDPC, Turbo, Polar**	Use when storage density is paramount. These codes offer near-capacity performance through iterative decoding (Soft-decision), though at the cost of higher computational latency.
3. Channels with Synchronization Drift	High Insertion/Deletion (Indel) rates; Loss of alignment	**Convolutional (HEDGES), Watermark Codes**	Crucial when the "position" of bits is uncertain. HEDGES/Convolutional codes use historical context to maintain synchronization without needing external alignment tools.
4. Large-Scale Data Pools (Unordered)	Packet Erasure (Missing DNA strands/Dropouts)	**Fountain Codes (LT, Raptor)**	Ideal for "DNA Soup" storage where entire oligonucleotides may be lost during PCR or sequencing. Being rateless, they allow recovery from any sufficient subset of retrieved strands.
5. Biochemical Stability Prioritized	Homopolymers (>3 identical bases); Extreme GC Content	**Constrained Codes (YYC), Hybrid Mapping**	Select these when synthesis or biological stability is the bottleneck. These act as "Source Coding" to prevent error-prone sequences from being generated in the first place.
6. Burst Error Mitigation	Localized, consecutive errors (e.g., strand breakage)	**Interleaved RS, Product Codes**	Standard block codes fail if errors exceed the correction capability of a single block. Interleaving distributes burst errors across multiple codewords, making them correctable.

strategies, achieving higher synthesis success rates (robustness) while maintaining high coding density (efficiency).

- **Run-length limitation**: q-ary system insertion/deletion correction codes [201] enforce strict maximum homopolymer run-lengths (<3–4 nucleotides) via synchronized markers, enhancing sequencing accuracy and overall robustness while providing single-edit error correction capability.
- **Secondary structures**: Direct construction methods [202] limit stem-loop lengths to avoid hairpins or dimers, while reversible/reversible-complement codes [203], [204] (inspired by Reed-Muller designs) completely eliminate secondary structures, maximizing robustness without sacrificing coding efficiency.

Other recent innovations include triple-constraint codes [205], [206] that simultaneously address secondary structures, run-length limits, and GC balance; capacity-approaching constructions [207] that achieve coding densities close to the theoretical limit (efficiency); and combinatorial designs [208] that enhance robustness via strict Hamming distance control and avoidance of reverse complements. Modern approaches [209], [210] further incorporate thermodynamic stability parameters, aligning the robustness design with the actual biochemical conditions.

D. NEW CODING SCHEMES FOR ACCESSIBLE DNA DATA STORAGE

DNA barcodes [152]—short nucleotide sequences used to uniquely index data fragments—directly correspond to the accessibility dimension in DNA coding, providing cost-effective random access for large-scale storage systems. However, barcodes face robustness challenges due to synthesis and sequencing errors, including substitutions, insertions, and deletions. Recent studies have made significant progress in improving both accessibility (accuracy) and robustness (fault tolerance).

- Early Hamming code–based barcodes [211] could only correct substitution errors, limiting robustness. Subsequently, Levenshtein codes [212], [213], [214] were developed to correct all three error types—substitutions, insertions, and deletions—substantially enhancing barcode robustness and thus improving accessibility.

- Press [215] demonstrated that long random barcodes ($\geq$34 nucleotides), combined with GPU-accelerated triplet decoding, can tolerate error rates exceeding 20%, significantly enhancing robustness while maintaining large-scale random access capability.
- Yu et al. [216] proposed a probability divergence–based barcode optimization method, achieving a 98.29% demultiplexing accuracy in nanopore sequencing data, balancing accessibility (high accuracy) and efficiency (avoiding excessively long barcodes).

This evolution—from rigid error-correcting codes to probabilistic frameworks—highlights the critical synergy between accessibility and robustness. Future research is likely to focus on integrating machine learning into barcode design to further expand the capacity and performance of barcode libraries.

IX. PERFORMANCE ASSESSMENT

The performance of DNA storage encoding schemes is typically evaluated through a combination of computational simulations, theoretical analysis, and experimental validation. Computational simulations model the entire DNA storage pipeline, incorporating the effects of synthesis, storage, and sequencing errors to assess the robustness of encoding schemes. Theoretical analysis provides capacity bounds by considering biochemical constraints such as GC content balance, homopolymer length, and sequence compatibility, offering insights into coding efficiency and design limits. Experimental validation complements these methods by quantifying real-world performance metrics such as information density, synthesis fidelity, and decoding accuracy. Together, these three approaches provide a comprehensive framework for evaluating DNA storage systems, enabling fair performance comparison, guiding parameter optimization, and supporting the development of reliable, high-capacity storage architectures. This section focuses on the two primary in silico evaluation methods: computational simulations and theoretical analysis.

A. CHANNEL MODELING AND SIMULATION PLATFORMS

The development of robust DNA storage systems is highly dependent on integrated channel modeling and simulation frameworks that capture the unique properties of molecular data storage. One widely adopted approach models DNA storage as a multi-subchannel system, where input sequences generate Poisson-distributed output clusters through replication processes. In this context, the noisy shuffling-sampling channel model [217] effectively captures critical characteristics such as molecular randomness, short sequence lengths, and base-level errors. Further refinement is provided by specialized nanopore sequencing models [150], [218], [219], which account for sequencing platform–specific error patterns and structural biases. Complementing these theoretical models, simulation platforms such as Chamaeleo [220] and other open-source toolkits [221], [222], [223], [224] offer end-to-end simulation capabilities. These tools integrate encoding and decoding algorithms with biochemical process modeling, including DNA synthesis, PCR amplification, sequencing, and error injection/correction mechanisms. This combined modeling–simulation paradigm facilitates predictive performance evaluation, supports the optimization of encoding strategies, and enables cost-effective pre-experimental validation. By allowing researchers to test and refine molecular storage architectures in silico, such tools significantly accelerate the design and deployment of practical DNA storage solutions.

B. CHANNEL CAPACITY

Channel capacity is a fundamental performance metric in DNA data storage, representing the theoretical upper bound on the rate at which information can be reliably encoded and retrieved in the presence of noise and biochemical errors. It provides a critical benchmark for evaluating the efficiency of various coding and decoding schemes under practical constraints such as synthesis imperfections, sequence degradation, and sequencing inaccuracies. Over the past few years, significant theoretical advancements have been made. Lenz et al. laid important groundwork by deriving upper bounds under substitution error models [225], analyzing achievable rates using concatenated coding architectures [226], and assessing capacity under suboptimal decoding conditions [227]. Shomorony and Heckel [217] expanded this analysis by incorporating different sampling distributions and base error models, while Weinberger and Merhav [228] refined capacity estimates by including insertion and deletion errors. From a data security perspective, Vippathalla and Kashyap [229] introduced a secure storage capacity framework that considers access control and confidentiality. At a more systemic level, [230] modeled DNA storage channels as cascades of noisy synchronization and shuffling-sampling subchannels, yielding tighter and more realistic bounds. In addition, [231] proposed a simplified computational framework for analyzing capacity under biochemical constraints such as run-length limitation (RLL) and GC content balance, and established conversion relationships between quaternary and binary constrained channels. Collectively, these studies provide a comprehensive theoretical foundation for understanding the limits of DNA storage systems and guide the development of more efficient and robust coding schemes.

C. KEY PERFORMANCE DIMENSIONS

1) INFORMATION DENSITY

Information density and coding efficiency are key performance metrics in DNA data storage, typically quantified by mass-based density (bits per gram) and nucleotide-based density (bits per nucleotide). While the theoretical upper limit for nucleotide-based efficiency is 2 bits per nucleotide, practical implementations face several challenges. Goldman et al. [160] estimated a mass-based storage potential of up to

2.2 petabytes per gram, but real-world systems must contend with error correction overhead, such as redundancy 20 to 30% from Reed-Solomon codes, and biochemical constraints, including homopolymer avoidance, GC content balance, and sequence complexity requirements. Grass et al. [6] demonstrated a trade-off strategy by using alternating nucleotide mappings to achieve 1.6 bits per nucleotide, sacrificing some efficiency for biochemical stability. Similarly, the DNA Fountain approach [122] reached 1.57 bits per nucleotide while ensuring strong error resilience. The ongoing research aims to bridge this gap by optimizing encoding methods that satisfy biochemical constraints without significantly compromising the information density.

2) BIOCHEMICAL CONSTRAINTS

DNA storage systems must operate within stringent biochemical stability constraints to ensure the reliability and longevity of stored data. These constraints are primarily governed by three key factors: GC content balance, sequence complexity, and structural stability. Church et al. [7] demonstrated that deviations from the optimal 50% GC content can significantly increase synthesis and sequencing error rates, with error levels rising 3–5 times in a 5.27 MB DNA storage experiment. In addition, maintaining high sequence complexity is crucial to the success of molecular processes such as PCR amplification. Organic et al. [152] found that optimizing k-mer entropy improved PCR survival rates from under 90% to over 99.9% by reducing repetitive sequence patterns like "CACACACA." Structural stability also plays a critical role, as thermodynamically unstable secondary structures can hinder synthesis and sequencing. Effective sequence design must therefore avoid stem-loop structures with stems shorter than three base pairs and ensure melting temperatures fall within acceptable ranges. These constraints necessitate the use of constrained coding techniques that integrate Shannon's information theory with Watson–Crick base-pairing rules, enabling the generation of sequences that are both information-dense and biochemically stable, thereby advancing the practical realization of DNA as a scalable and robust data storage medium.

3) ERROR CORRECTION CAPABILITY

The error correction capability of a DNA storage system is a critical factor determining data reliability and retrieval accuracy. During encoding, storage, and sequencing, DNA strands are vulnerable to various types of errors, including base substitutions (e.g., C→T transitions), insertions and deletions (indels), and strand breaks. The nature and frequency of these errors depend on the sequencing technology used: Illumina platforms generally show low substitution error rates between 0.1% and 1%, whereas long-read technologies such as PacBio exhibit higher indel rates ranging from 5% to 15% [232]. As discussed in previous sections, to ensure data integrity, a variety of error-correcting codes have been developed to address these diverse error types. For example, Reed–Solomon (RS) codes are widely adopted for their robustness against burst errors, capable of recovering up to 10% data loss with approximately 20% redundancy [6]. More advanced codes, such as low-density parity-check (LDPC) codes, provide near-capacity error correction performance and greater efficiency, especially suitable for large-scale datasets. The DNA Fountain approach [122], which incorporates LDPC coding, demonstrated complete data recovery from a 1.6 MB dataset despite a fragment loss rate of 15%, exemplifying an effective trade-off between correction strength and redundancy overhead. Ultimately, the optimal error correction strategy should be tailored to the specific error characteristics of the system and the requirements of the intended application.

4) TRADE-OFF ANALYSIS

A central challenge in the development of DNA storage systems is achieving a balanced trade-off among information density, error correction reliability, and biochemical compatibility. Different application scenarios, such as long-term archival storage and rapid data retrieval, impose varying demands on encoding schemes, requiring customized optimization strategies. For example, the DNA Fountain method [122] demonstrates a favorable balance by achieving a high storage density of approximately 1.57 bits per base pair, while tolerating up to 15% data loss and allowing near-perfect recovery. In contrast, simple redundancy schemes like triple replication can reduce error rates to below 10^{-6} [95] but at the cost of significantly lower information density—around 0.67 bits per base pair—highlighting the inherent trade-off between error correction capability and storage efficiency. This comparison underscores the need for more advanced coding techniques. To enable objective evaluation and optimization of DNA storage technologies, standardized benchmarking tools and comprehensive evaluation frameworks, combined with theoretical modeling, are essential. As these methodologies continue to improve, they will play a pivotal role in advancing the maturity and commercialization of DNA-based data storage systems.

X. FUTURE RESEARCH DIRECTIONS

The advancement of DNA data storage technology is entering a critical phase, necessitating the evolution of encoding methodologies to overcome fundamental limitations and address practical implementation challenges. Future research should focus on several key areas that include theoretical foundations, computational algorithms, and applied engineering to drive the technology toward broader feasibility and adoption.

A. ADVANCED ERROR MODELING AND ADAPTIVE ENCODING STRATEGIES

The development of next-generation DNA storage systems requires more refined approaches to error modeling and mitigation. Traditional uniform error assumptions are increasingly inadequate given the platform-specific and sequence-dependent nature of synthesis and sequencing errors. Recent progress in hierarchical Bayesian modeling [233] has shown great potential to capture complex and heterogeneous error patterns across different DNA synthesis platforms. By integrating these models with dynamic redundancy allocation strategies, it is possible to enhance both data integrity and storage efficiency. A promising direction for future research lies in the design of context-aware encoding schemes that dynamically adjust error correction parameters in response to real-time analysis of sequence-specific error probabilities. Such adaptive encoding mechanisms offer the potential to transcend the conventional trade-offs between reliability and storage density, paving the way for more robust and efficient DNA-based information systems.

B. COMPUTATIONAL ACCELERATION VIA PARALLEL AND EMERGING ARCHITECTURES

The high computational complexity of DNA encoding and decoding remains a significant barrier to the practical deployment of large-scale DNA storage systems. To address this challenge, recent efforts have focused on leveraging parallel computing architectures, particularly GPU-accelerated pipelines, which have demonstrated order-of-magnitude improvements in processing speed for large-scale data conversion tasks. These enhancements significantly reduce the time required for sequence generation, constraint validation, and error correction. In addition to traditional parallelization, emerging approaches—such as quantum-inspired optimization algorithms—offer promising avenues to efficiently solve complex constraint satisfaction problems associated with DNA code design. These algorithms can explore vast solution spaces more effectively than classical methods, potentially enabling faster and more scalable encoding strategies. Continued advances in computational architectures will be critical to supporting the real-time processing demands of commercial-scale DNA storage applications.

C. CROSS-DISCIPLINARY SYNERGIES FOR ENCODING INNOVATION

The most impactful advances in DNA storage encoding are emerging increasingly from the integration of insights across multiple disciplines. Computer science plays a pivotal role through the development of constrained coding theories tailored to the biochemical and structural constraints of DNA molecules. At the same time, molecular biology introduces groundbreaking possibilities, such as the use of CRISPR-Cas systems for in vivo error detection and correction [234], which may redefine how biological mechanisms contribute to data reliability. In parallel, materials science supports practical deployment by designing advanced stabilization matrices that enhance DNA durability while enabling efficient random access. This convergence of computational, biological, and material expertise fosters a multidisciplinary framework for developing encoding strategies that are not only theoretically sound but also practically implementable, accelerating the translation of DNA storage from research to real-world applications.

D. ENCODING STRATEGIES FOR ENVIRONMENTAL RESILIENCE

As DNA data storage advances toward practical deployment, encoding strategies must be engineered to withstand a wide range of environmental stressors. In space-based applications, exposure to high levels of radiation necessitates the development of radiation-tolerant codes capable of mitigating damage from cosmic rays and ionizing particles. For deep-sea storage, encoding schemes must ensure hydrothermal stability by incorporating thermodynamically robust sequences that can endure high pressure and variable temperatures. Even terrestrial environments present challenges such as oxidative degradation, which can be addressed through the design of protective sequence motifs that enhance chemical stability. Critically, these environmental resilience factors should be integrated into the core design of encoding algorithms rather than treated as afterthoughts or external constraints. Embedding environmental robustness at the algorithmic level will be essential to ensure the long-term reliability and viability of DNA storage across diverse application domains.

E. TOWARD A STANDARDIZATION FRAMEWORK FOR SCALABLE DEPLOYMENT

The successful transition of DNA storage from experimental research to commercial-scale systems hinges on the establishment of comprehensive standardization frameworks. These frameworks must encompass interoperability protocols that enable seamless integration across diverse synthesis and sequencing platforms, thereby ensuring system compatibility and flexibility. Equally important is the development of robust benchmarking metrics that facilitate the objective evaluation of the performance of encoding schemes under consistent and reproducible conditions. In addition, the field must adopt modular architectural specifications that support iterative innovation while preserving backward compatibility, enabling ongoing advancements without compromising existing systems. Such coordinated standardization efforts are essential to foster industry-wide collaboration, accelerate technological maturation, and lay the foundation for DNA storage to become a scalable, sustainable, and reliable solution for long-term digital archiving.

F. MACHINE LEARNING APPROACHES FOR DNA STORAGE CODING

In recent years, machine learning (ML) has become an important driving force to overcome bottlenecks in DNA data storage coding. However, this field still faces three major challenges. First, model training is highly dependent on synthetically generated data, and validation under real biological experimental conditions remains insufficient [235]. Second, most existing approaches are optimized for specific data types, leaving their general applicability to various storage scenarios to be further improved [236]. Finally, the "black-box" nature of neural networks limits the interpretability of the encoding process, making it difficult to meet biological safety and regulatory requirements. Future research should therefore focus on the development of cross-modal pretraining architectures, lightweight model deployment strategies, and explicit biological constraint embedding mechanisms, while strengthening collaboration between academia and industry to promote the establishment of standardized evaluation frameworks.

XI. CONCLUSION

In this paper, we have systematically explored the trajectory of DNA-based data storage, moving from its molecular foundations—which establish DNA as a dense, durable, and replicable medium—to the intricate biochemical constraints that differentiate this organic substrate from conventional silicon-based systems. By reviewing the complete workflow including synthesis, storage, and sequencing, we have identified a distinct methodological convergence in recent literature: the shift from adapting rigid, classical error-correcting codes to developing flexible, biochemically-aware encoding schemes.

Our analysis reveals a common view among researchers that while classical adaptations (e.g., Reed–Solomon, BCH) and modern probabilistic schemes (e.g., Fountain, LDPC codes) provide foundational reliability, they are increasingly being superseded or augmented by domain-specific approaches. The review of cutting-edge strategies—such as Damerau metric, Asymmetric Lee Distance (ALD), set-based, and Yin-Yang codes—demonstrates a consensus that optimal storage efficiency cannot be achieved without intrinsic compliance with biochemical constraints (e.g., GC-content balance and homopolymer avoidance) and robustness against insertion/deletion errors.

Furthermore, a synthesis of current evaluation practices highlights a critical reliance on both theoretical channel models and system-level simulations. Our performance comparisons indicate that while current benchmarks successfully quantify the trade-offs between redundancy and error resilience, there remains a disconnect between theoretical capacity and wet-lab viability. Consequently, the choice of encoding scheme is not merely a mathematical optimization but a determinant factor in the system's physical scalability and cost-efficiency.

Looking forward, the implications of these findings suggest that future research must move beyond isolated coding designs. We identify the following priority directions:

- Adaptive Methodologies: The development of adaptive encoding methods tailored to dynamic error profiles found in emerging synthesis and sequencing technologies.
- Interdisciplinary Integration: Intelligent designs that leverage machine learning and molecular computing to predict and mitigate biochemical noise.
- Standardization: The establishment of standardized benchmarks and ethical frameworks, which are essential to transition DNA storage from experimental proof-of-concepts to a safe, scalable, and commercially viable solution for next-generation digital archiving.

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